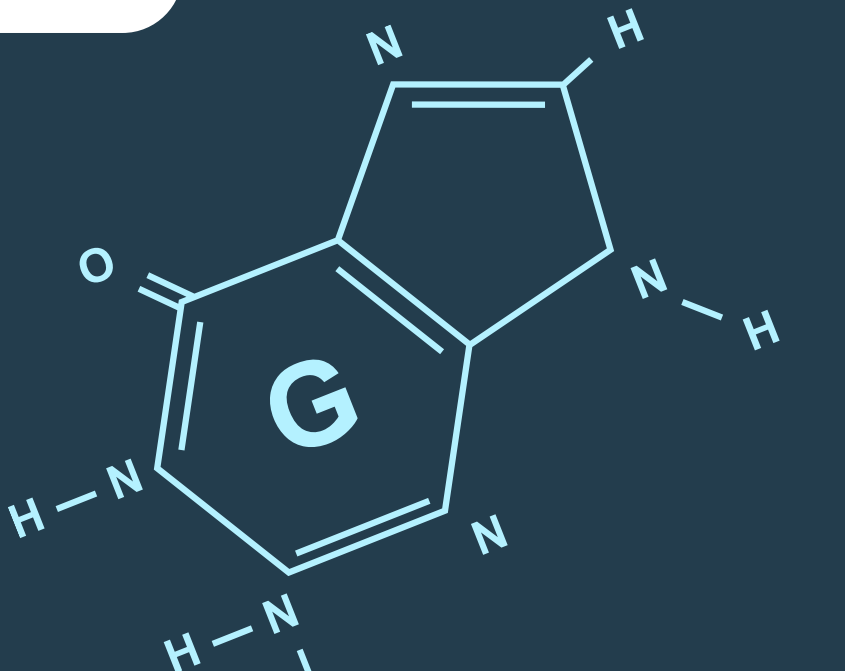
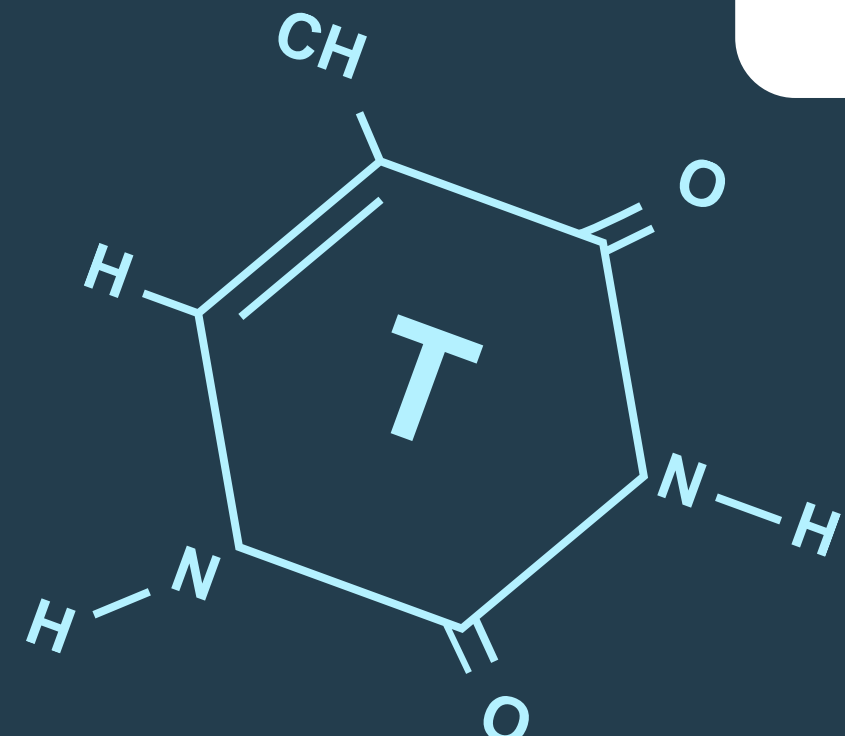
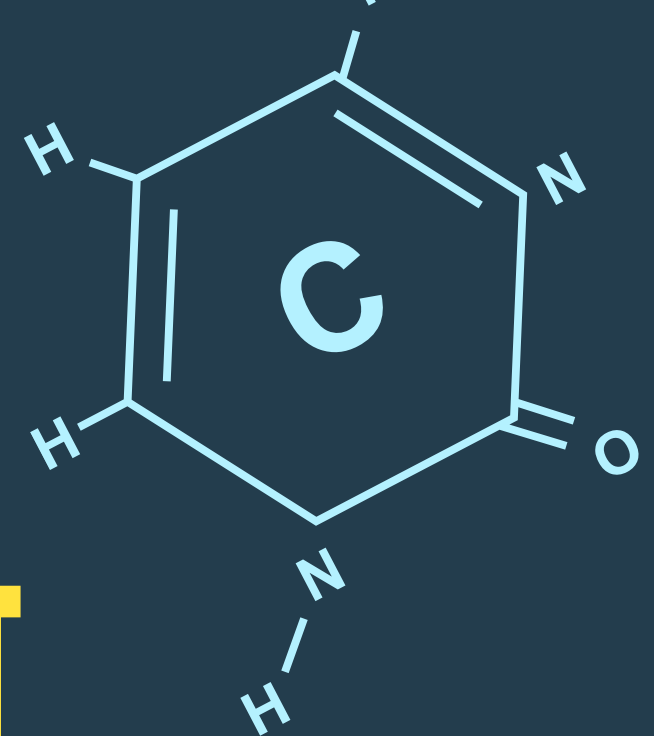
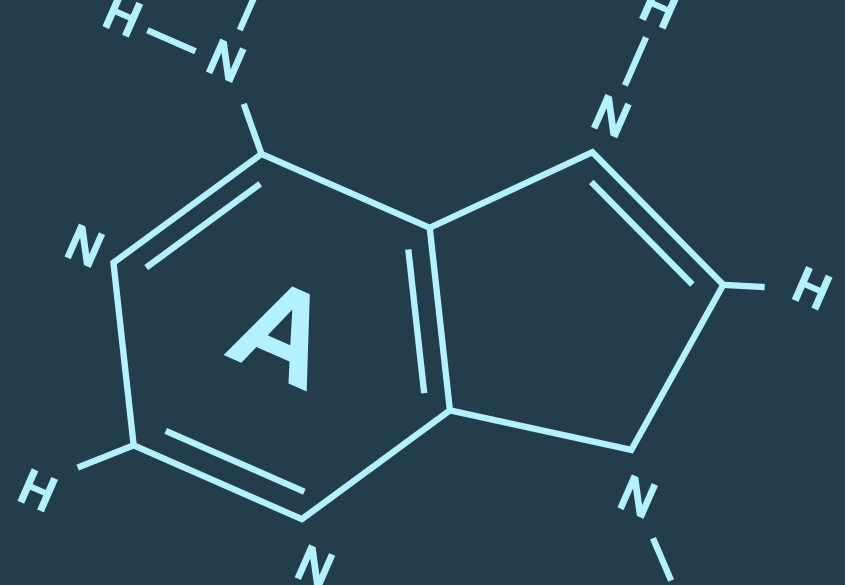




TARA CUDA: A HYBRID SIMT IMPLEMENTATION OF ZHANG'S SIX- FRAME ALGORITHM USING CUDA

ARCETA - MENDOZA - TAN

ADVISER: MR. ROGER LUIS UY

BACKGROUND

Zhang's six-frame alignment

- **DNA-protein alignment** is a type of sequence alignment used to identify similarities between DNA and protein sequences.



Zhang's six-frame alignment

DNA SEQUENCE INPUT

ATGCACTGC...



Zhang's six-frame alignment

ATGCACTGC...



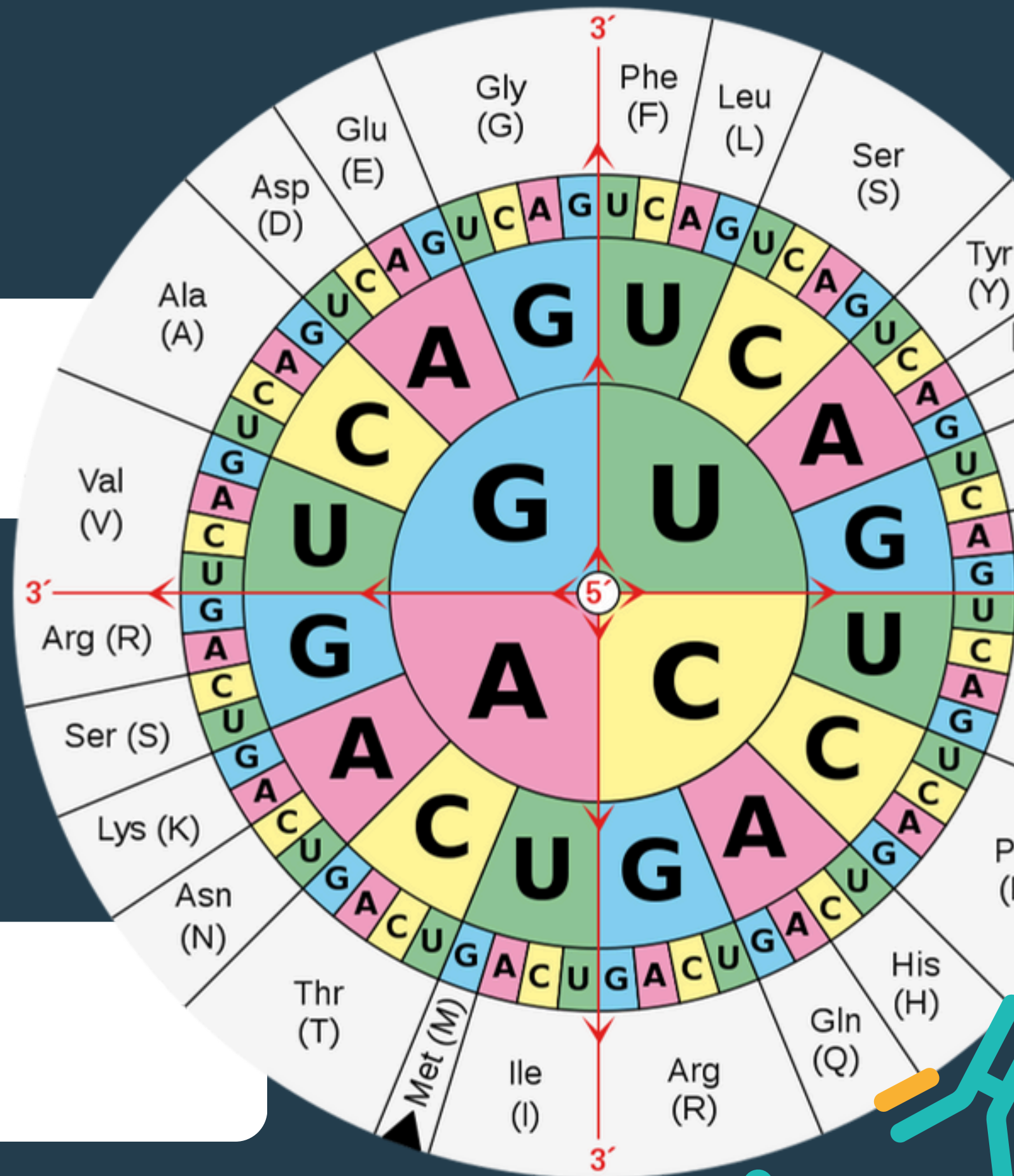
F1

M

H

C

...



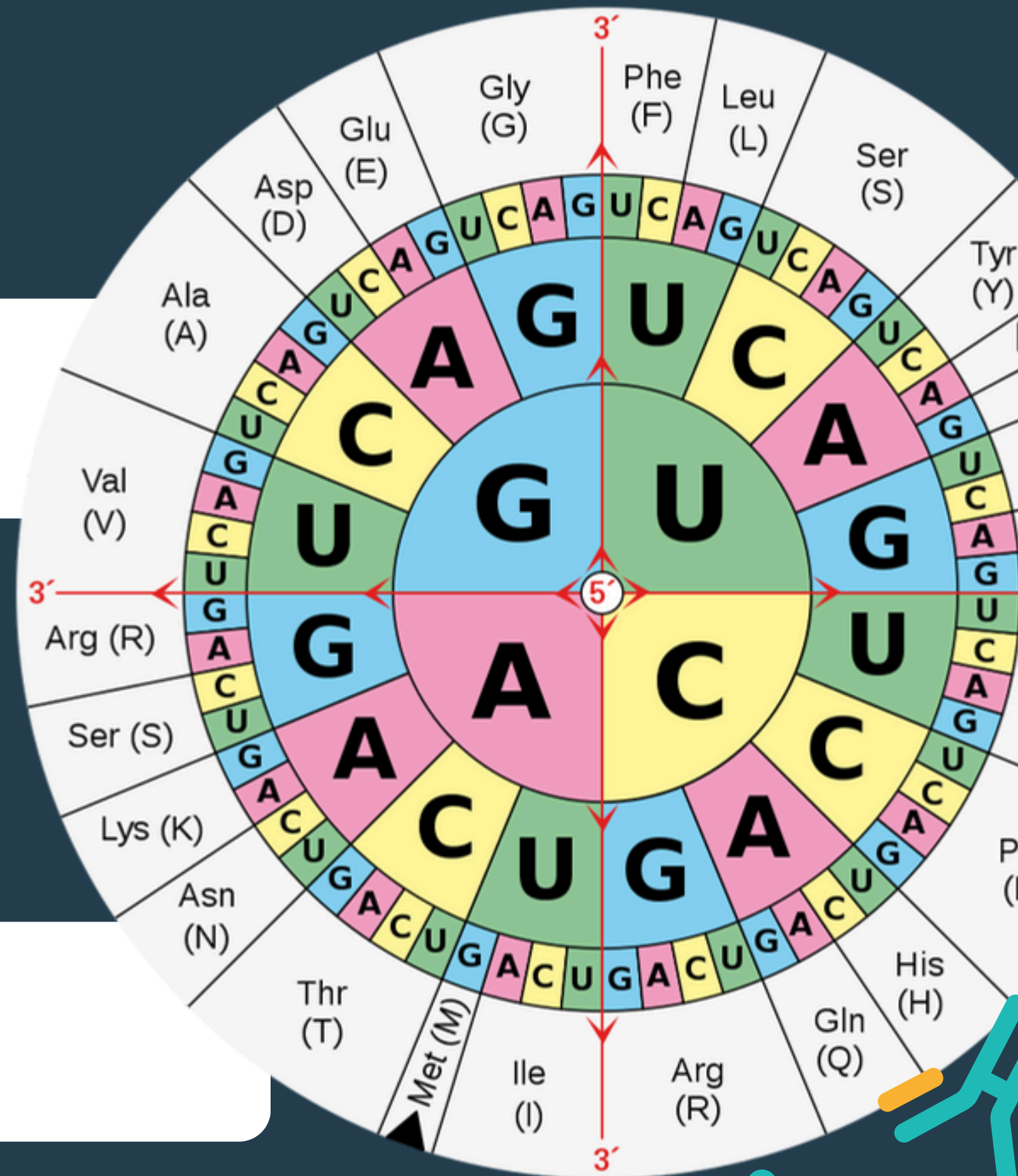
Zhang's six-frame alignment

ATGCACTGC...



F2

C T ...



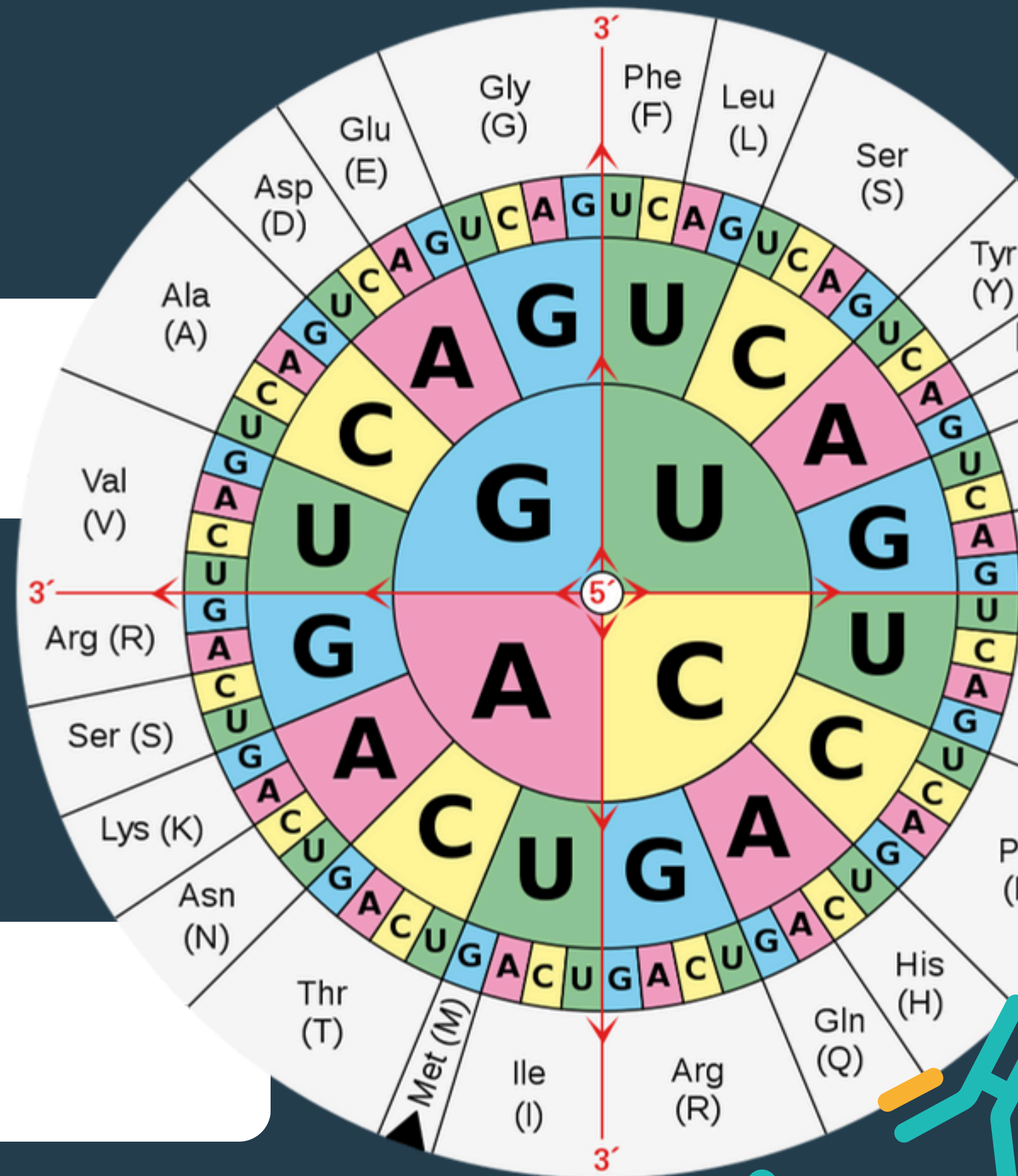
Zhang's six-frame alignment

ATGCACTGC...



F3

AL...



ATGCACTGC...

F1

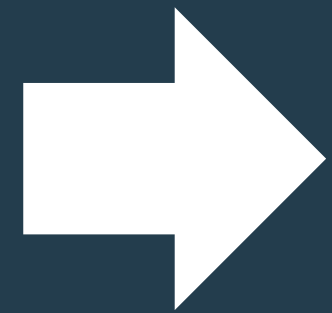
M H C ...

F2

C T ...

F3

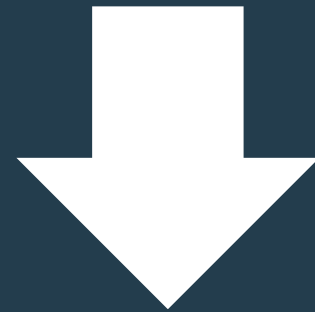
A L ...





Zhang's six-frame alignment

ATGCACTGC...



AGTGTGCAC...

*reverse complement



AGTGTGCAC...

F4

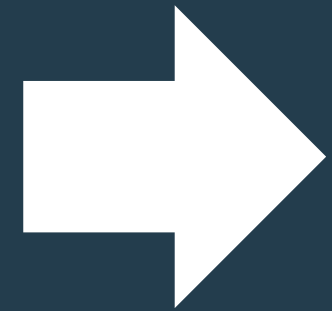
S V H ...

F5

V C ...

F6

C A ...





BACKGROUND



Zhang's six-frame alignment

- **DNA-protein alignment** is a type of sequence alignment used to identify similarities between DNA and protein sequences.
- While **Zhang's six-frame alignment algorithm** is an established method, its sequential implementation can be computationally expensive, particularly when dealing with large datasets.

Hence, to address this issue, **parallel techniques** such as **SIMT (Single Instruction, Multiple Threads)** and **SIMD (Single Instruction, Multiple Data)** are used to enhance the execution of these algorithms.





GENERAL OBJECTIVE

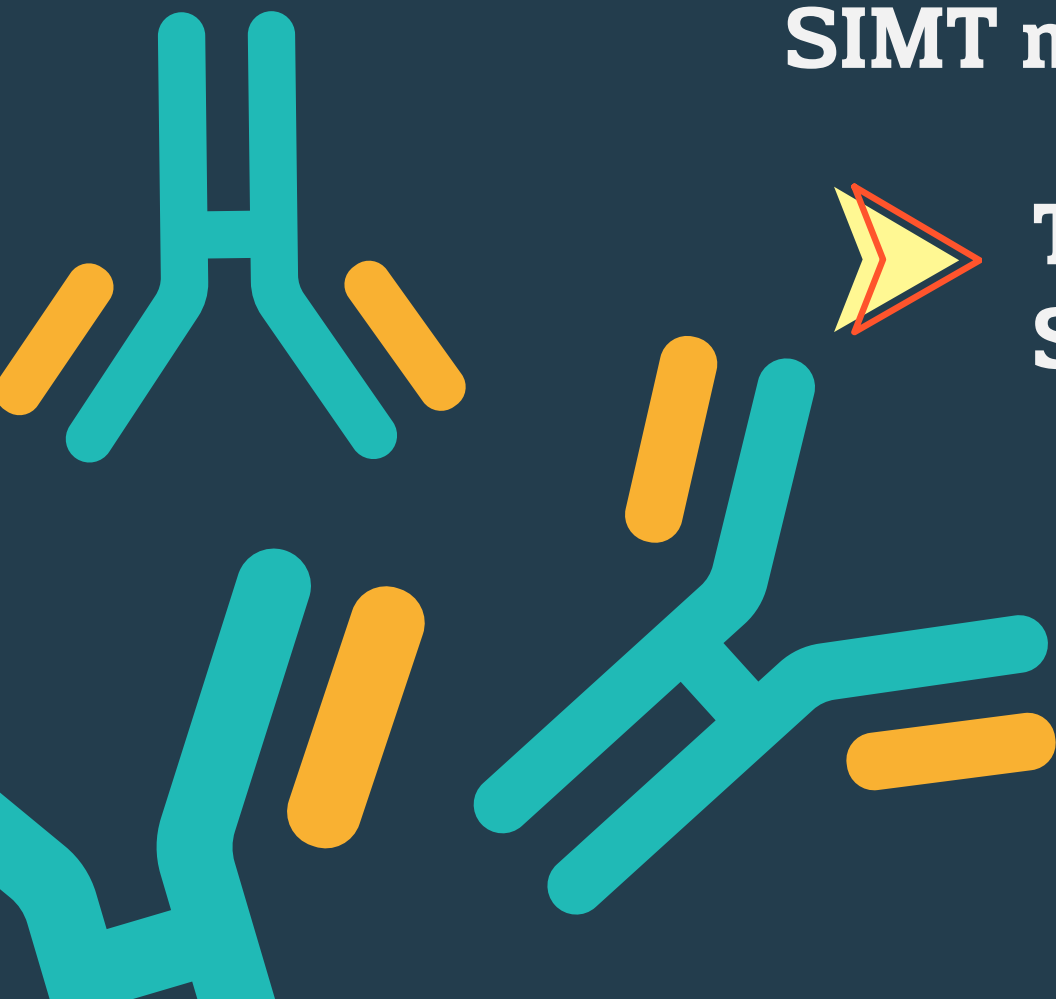
This research aims to develop an approach to implementing Zhang's Six-Frame Alignment Algorithm in parallel, specifically using the CUDA SIMT execution model.



SPECIFIC OBJECTIVES



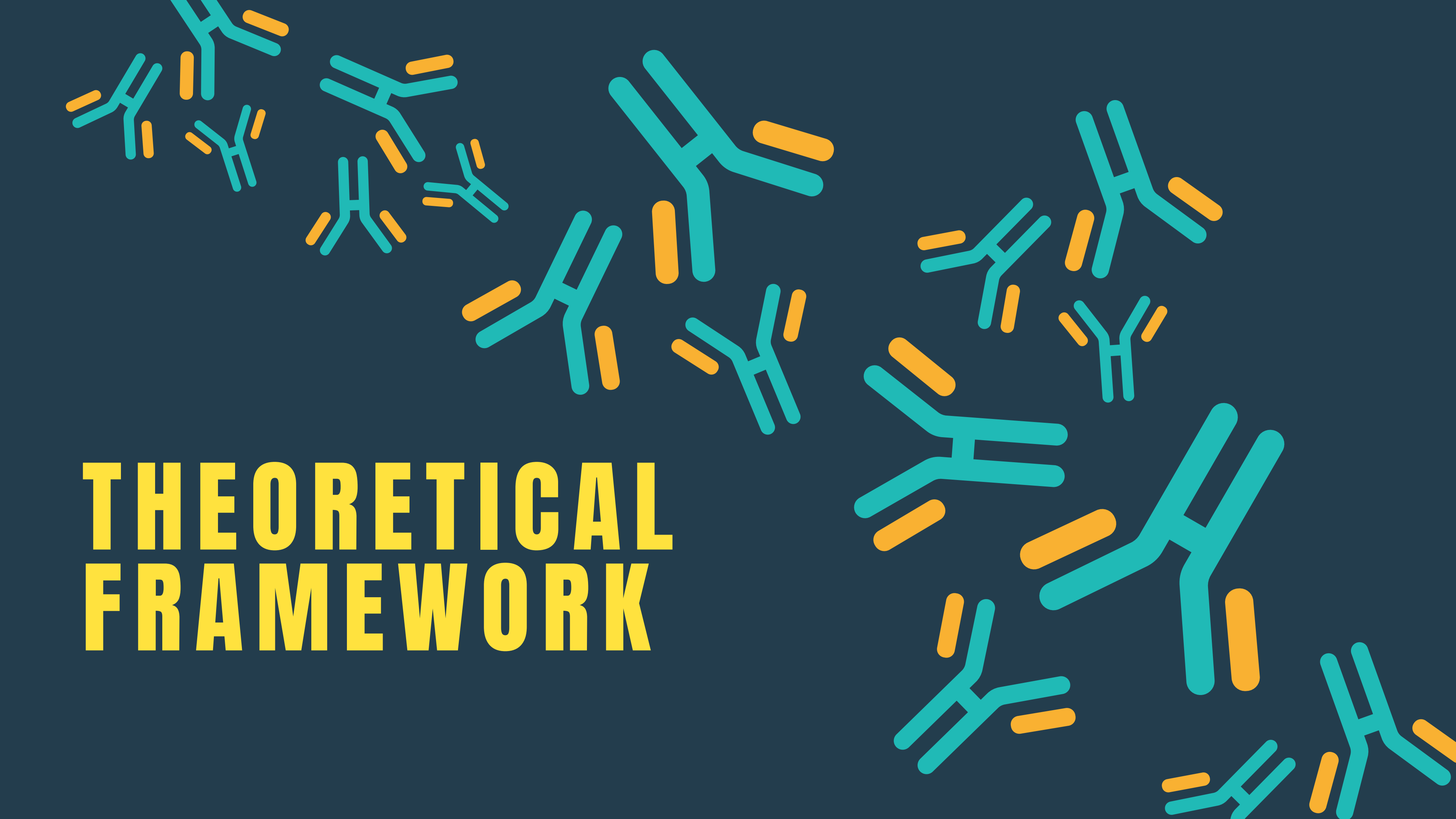
- To **develop** a sequential implementation of Zhang's Six-Frame Alignment Algorithm.
- To **create** an implementation scheme for Zhang's Six-Frame Alignment Algorithm to better fit the parallel framework of the SIMT model.
- To **implement** Zhang's Six-Frame Alignment Algorithm with the SIMT execution model using CUDA.
- To **evaluate** the developed parallel implementation by comparing and contrasting its performance to the sequential version and existing SIMD implementation.





SCOPE AND LIMITATIONS

- This research mainly focuses on **DNA to Protein** Sequence Alignment. The algorithm to be implemented is **Zhang's six-frame algorithm**.
 - While many parallelization methods exist, the study will focus only on the **SIMT execution model using CUDA**.
 - The proposed parallel implementation of this research will mainly focus on the **scoring part** of Zhang's six-frame algorithm.
 - The proposed implementation will be evaluated and compared with the **sequential** and **previous parallel implementations**.
- 

The background of the entire image is a solid dark blue. Scattered across this background are numerous abstract, branching, Y-shaped or T-shaped structures. These structures are composed of thick teal lines. Attached to the ends of these teal branches are short, thick orange lines, some of which are oriented horizontally and others at an angle. The overall effect is a dense, organic, and somewhat chaotic pattern of these teal and orange elements.

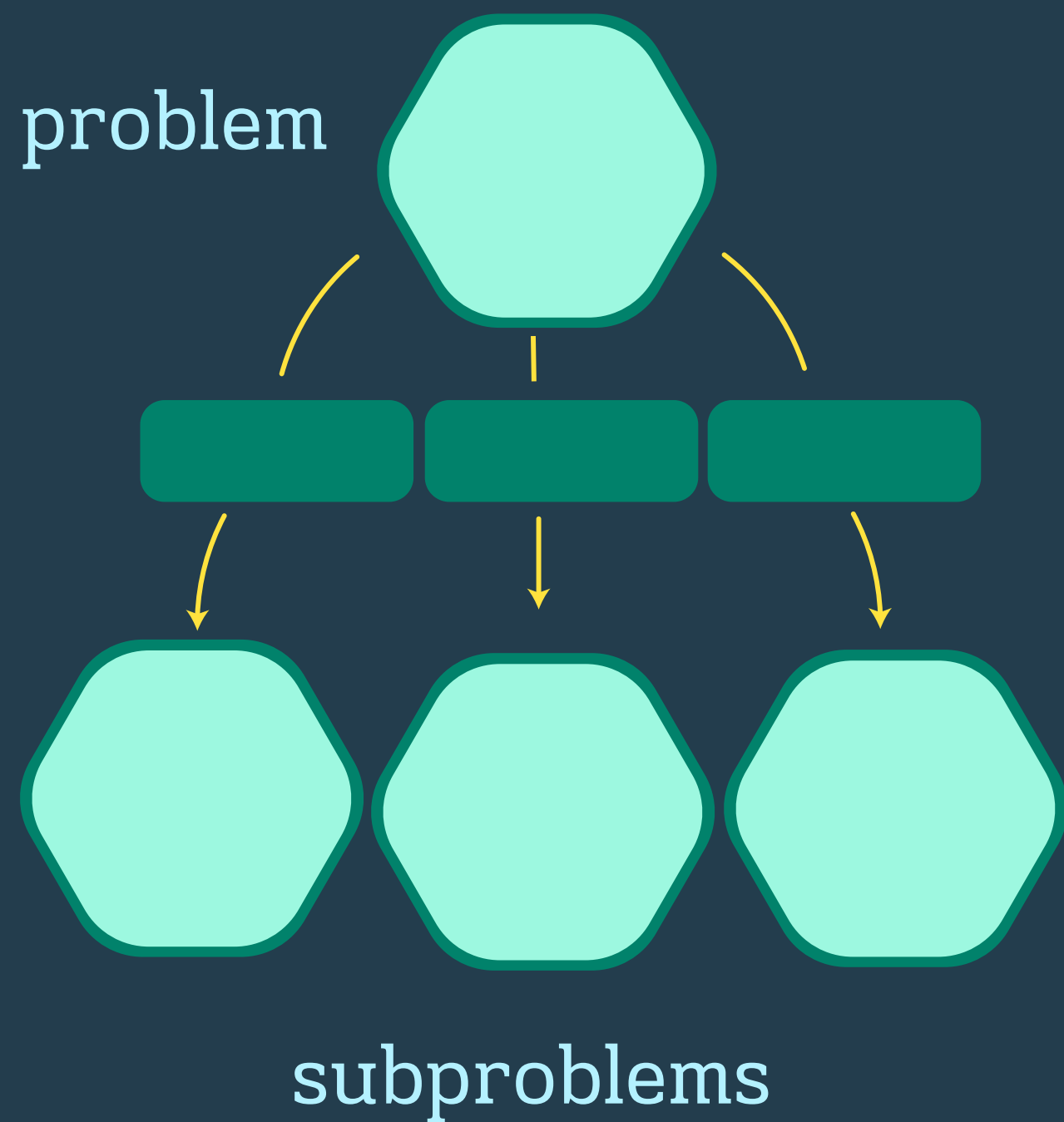
THEORETICAL FRAMEWORK

CUDA & SIMT PARADIGM

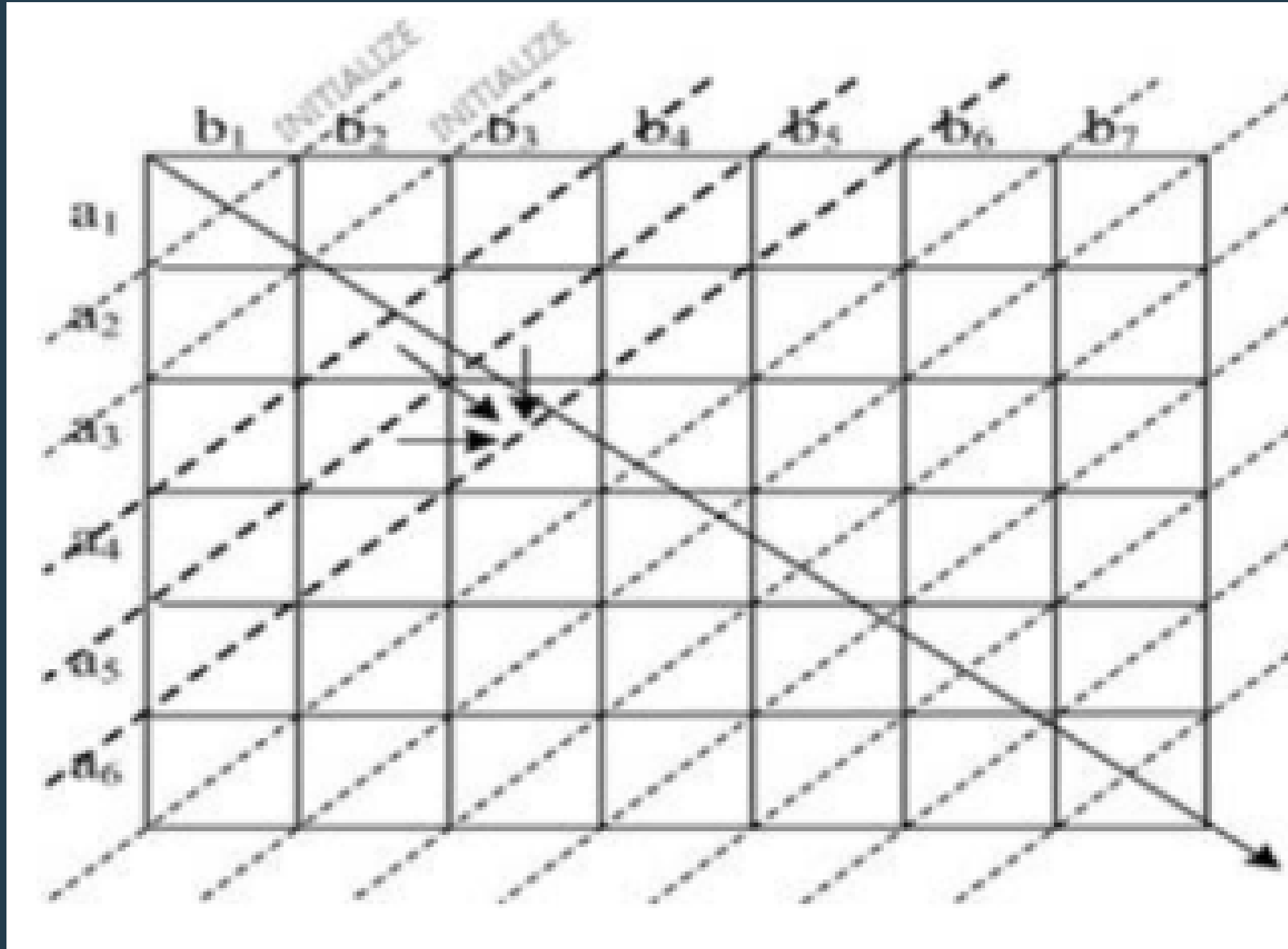


- CUDA's interface **enables high-performance computing tasks** that would otherwise be inefficient in traditional CPU implementations.
 - CUDA's design is based on the **SIMT approach**, which allows several threads to execute identical instructions on different data streams, resulting in a massively parallel computation.
- 

CUDA & SIMT PARADIGM



To harness the parallelism capabilities of CUDA, the problem must be **broken down** into independent tasks that can be performed concurrently. This breakdown makes it possible for many threads to be executed at the same time, which is the main benefit of CUDA.



The **Anti-Diagonal Technique** is a technique used for filling in a matrix by processing its elements along anti-diagonals instead of the traditional row-by-row or column-by-column approaches. This enables parallelization for algorithms with top, left, and top-left dependencies.

WAVEFRONT PARADIGM



SYSTEM IMPLEMENTATION

- The datasets utilized in this research comprise protein and DNA sequences for *Drosophila melanogaster* (fruit fly) and *Arabidopsis thaliana* (mouse-ear cress).
- Scraped via Selenium from UniProt & European Nucleotide Archive (ENA) website

UniProt

BLAST

Align

Peptide search

ID mapping

SPARQL

Proteomes

Advanced

List

Search

Proteomes · Drosophila melanogaster (Fruit fly)

Proteome IDⁱ

UP000000803

Status

Reference proteome

Number of entriesⁱ

22,032

Gene countⁱ

13,823

Download one protein sequence per gene (FASTA)

Taxonomy | Strain

Drosophila melanogaster (Fruit fly) | Berkeley

Genome representation

Full

Completeness (CPD)ⁱ

Close to standard (high value)

BUSCOⁱ

Single

Duplicated

Fragmented

Missing

n:3285 · diptera_odb10

Proteomes · Arabidopsis thaliana (Mouse-ear cress)

Proteome IDⁱ

UP000006548

Status

Reference proteome

Number of entriesⁱ

39,276

Gene countⁱ

27,448

Download one protein sequence per gene (FASTA)

Taxonomy | Strain

Arabidopsis thaliana (Mouse-ear cress) | cv. Columbia

Taxon ID

3702

Genome representation

Full

Completeness (CPD)ⁱ

Unknown

BUSCOⁱ

Single

Duplicated

Fragmented

Missing

n:4596 · brassicales_odb10

C:98.6% (S:63.9% D:34.7%) F:0.2% M:1.2%

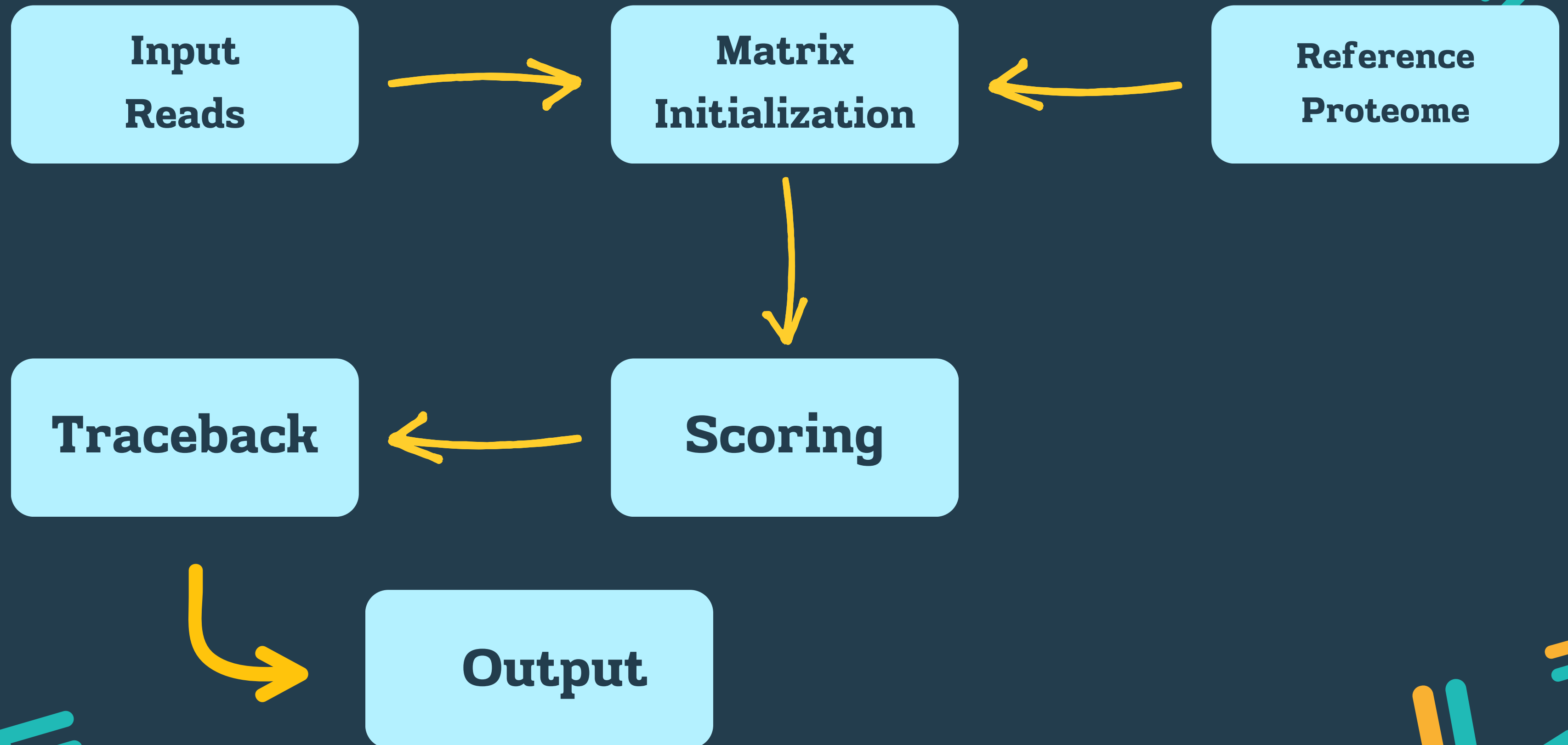
DATASET

- After preprocessing, the final datasets that will be used consist of valid and fully aligned protein and DNA sequence pairs for *Drosophila melanogaster* (fruit fly) and *Arabidopsis thaliana* (mouse-ear cress).
- *Drosophila melanogaster*
 - Dataset: 21,449 protein sequences (each with DNA translation)
 - Protein lengths: 11 bp (shortest) to 22,949 bp (longest)
 - Average protein length: ~451 bp
- *Arabidopsis thaliana*
 - Dataset: 39,067 protein sequences (higher count)
 - Protein lengths: 5 bp (shortest) to 5,400 bp (longest)
 - Average protein length: ~396 bp

DATASET



TARA CUDA IMPLEMENTATION



TARA CUDA IMPLEMENTATION

**Input
Reads**

GACGCTGTAGGCCATA...

TARA CUDA IMPLEMENTATION

**Input
Reads**



**Matrix
Initialization**



**Reference
Proteome**

TARA CUDA IMPLEMENTATION

```
ALL_proteom...000803.fasta  BLOSUM62.txt  seq.cpp
36509 TCIGGQTFKILTWLGYSNSAFNPITISIFNKEFRDAFRILTMRNPWCCAQDVGNIHPRN
36510 SDRFITDYAAKNVVMNSGRSSAELEQVSAI
36511 >sp|P41894|DEI_DROME Helix-loop-helix protein delilah OS=Drosophila melanogaster OX=7227 GN=tx PE=1 SV=3
36512 MKSNTYELHNYADLNDMARATDSKSRKRKTASARGEKYSLRQKRQKRGSNEDGESANLA
36513 DFQLELDPIAEPASKSRKNAPTkskTKAPPLSKYRRKTANARERTRMREINTAFETLRHC
36514 VPEAIKGEDAANTNEKLTKITTLRLAMKYITMLTDSIRDPSYESEFIGECLEESANREAR
36515 VDLEANEAEVELPVPVAKKPAKTGSGKKSSAASKRQSQKQAKIVPQIPPISSGESCYA
36516 TSSIGSPASSAYASLSSSSNSHSSSSSPGLELDSLVLNGISALDSLILDTSDGDSL SCL
36517 SPGYGSLLTGSGESLLPGCRGDLMSGLEKSDVELSLRLLDQSSKDSFDFASDQQPSACI
36518 SSFSALDGYPFDLLHSDFPDMFLT
36519 >sp|P41964|DMYC_DROME Drosomycin OS=Drosophila melanogaster OX=7227 GN=Drs PE=1 SV=1
36520 MMQIKYLFALFAVLMLVVLGANEADADCLSGRYKGPCAVWDNETCRRVCKEEGRSSGHCS
36521 PSLKWCCEGC
36522 >sp|P42003|MAD_DROME Protein mothers against dpp OS=Drosophila melanogaster OX=7227 GN=Mad PE=1 SV=1
36523 MDTDDVESNTSSAMSTLGSLFSFTSPAVKKLLGWKQGDEEEKWAEKAVDSL VKKLKKRKG
36524 AIEELERALSCPGQPSKCVTIPRSLDGRLQVSHRKLPHVIYCRVWRWPDLQSHHELKPL
36525 ELCQYPFSAKQKEVCINPYHYKRVESPVLPVLPVRHSEFAPGHSMQLQFNHVAEPSMPHN
36526 VSYNSSGFNHSLSTSNTSVGSPSSVNSNPNSPYDSL AGTPPPAYSPSEDGNSNPNPDGG
36527 QLLDAQMGDVAQVSYSEPAFWASIAYYELNCRVGEVFHCNNNSVIVDGF TNPSNNSDRCC
36528 LGQLSNVNRNSTIENTRRHIGKGVHLYYVTGEVYAECLSDSAIFVQSRNCNYHHGFHPST
36529 VCKIPPGCSLKIFNNQEFAQLLSQSVNNGFEAVYELTKMCTIRMSFVKGWGAEYHRQDVT
36530 STPCWIEIHLHGPLQMLDKVLTQMGSPHNAISSVS
36531 >sp|P42124|EZ_DROME Histone-lysine N-methyltransferase E(z) OS=Drosophila melanogaster OX=7227 GN=E(z) PE=1 SV=2
36532 MNSTKVPEWKRRVKSEYIKIRQQKRYKRADEIKEAWIRNWEHNNHVQDLYCESKVWQA
36533 KPYDPPHVDCVKRAEVTSYNGIPSGPQKVPICVINAVTPIPTMYTWAPTQQNFMVEDET
36534 LHNIPYMGDEVLDKDGKFIELIKNYDGKVHGDKDP SFMDDAIFVELVHALMRSYSKELE
36535 EAAPGTATAIKTETLAKSKQGEDDGVVDVDADGESPMKLEKTD SKGDLTEVEKKETEEPL
36536 ETEDADV KPDVEEVKDKLPFPAPIIFQAISANFPDKGTAQELKEKYIELTEHQDPERPQE
36537 CTPNIDGIKAESVSRERTMHSFHTLFCRRCFKYDCFLHRLQGHAGPNLQKRRYPELKPFA
36538 EPCSNSCYMLIDGMKEKLAADSKTPPIDSCNEASSEDSDNSNSQFSNKDFNHENSKDNGL
```

**Reference
Proteome**

TARA CUDA IMPLEMENTATION

MITDVQLAIFSNVLGV...

**Reference
Proteome**



TARA CUDA IMPLEMENTATION

Matrix Initialization

- Initialize the first few columns and rows

Matrix I

Matrix D

Matrix C

TARA CUDA IMPLEMENTATION

Matrix Initialization

- Initialize the first few columns and rows

Matrix I

Mat

$$I(j, 0) = -\infty$$

$$D(0, j) = D(2, j) = D(3, j) = -\infty$$

$$D(1, j) = C(0, j) - gop - gep$$

$$C(0, 0) = 0$$

$$C(0, j) = I(0, j)$$

$$C(j, 0) = D(j, 0)$$

$$C(1, j) = \max \begin{cases} I(1, j) \\ D(1, j) \\ C(0, j-1) + S[b_1, a_j] \end{cases}$$

$$C(2, j) = \max \begin{cases} I(2, j) \\ C(0, j-1) + S[b_1, a_j] - \pi \end{cases}$$

$$C(3, j) = \max \begin{cases} I(3, j) \\ C(1, j-1) + S[b_2, a_j] - \pi \end{cases}$$

$$C(4, j) = \max \begin{cases} I(4, j) \\ D(4, j) \\ C(1, j-1) + S[b_4, a_j] \\ C(2, j-1) + S[b_4, a_j] - \pi \end{cases}$$

Matrix C

TARA CUDA IMPLEMENTATION

Matrix Initialization

- Initialize the first few columns and rows

$-\infty$	-5	-5	-5	-5	-5	-5
$-\infty$	-5	0	-2	-4	-5	-4
$-\infty$	-5	-5	-5	-5	-5	-5
$-\infty$	-5	-5	-5	-5	-3	-5
$-\infty$						

Matrix I

$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
-5	-5	-5	-5	-5	-5	-5
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						

Matrix C

TARA CUDA IMPLEMENTATION

Input



**Matrix
Initialization**



**Reference
Proteome**



Scoring



TARA CUDA IMPLEMENTATION



Scoring

$-\infty$	-5	-5	-5	-5	-5	-5
$-\infty$	-5	0	-2	-4	-5	-4
$-\infty$	-5	-5	-5	-5	-5	-5
$-\infty$	-5	-5	-5	-5	-3	-5
$-\infty$						

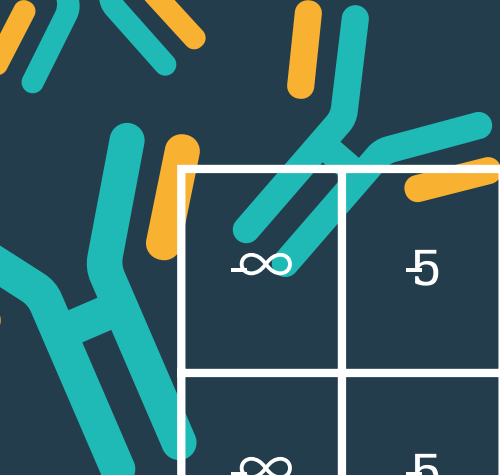
Matrix I

$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
-5	-5	-5	-5	-5	-5	-5
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						

Matrix C



∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞						
∞						
∞						
∞						
∞						

Matrix I

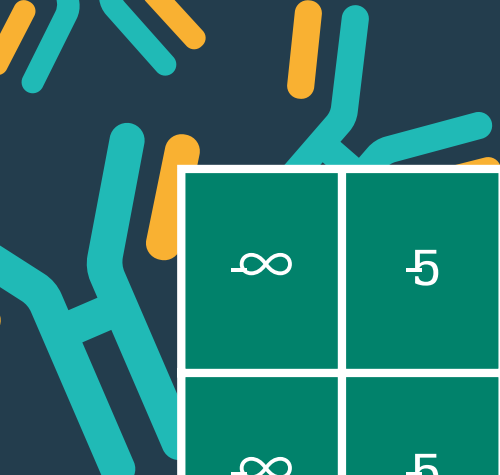
∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5						
5						
5						
5						
5						

Matrix D



0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C



∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞						
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5						
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C



∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞						
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	$I(i,j) = \max \begin{cases} I(i,j-1) - gep \\ C(i,j-1) - gep - gop \end{cases}$ $D(i,j) = \max \begin{cases} D(i-3,j) - gep \\ C(i-3,j) - gep - gop \end{cases}$ $C(i,j) = \max \begin{cases} I(i,j) \\ D(i,j) \\ C(i-2,j-1) + S[b_i, a_j] - \pi \\ C(i-3,j-1) + S[b_i, a_j] \\ C(i-4,j-1) + S[b_i, a_j] - \pi \end{cases}$					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C



TARA CUDA IMPLEMENTATION

Scoring

This process has 2 options since the program will prompt the user to choose between the **Sequential** or **CUDA (Parallel)** implementation of the selected algorithm.





TARA CUDA IMPLEMENTATION

Scoring

This process has 2 options since the program will prompt the user to choose between the **Sequential or CUDA (Parallel) implementation of the selected algorithm.**



Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5						
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

$$I(i,j) = \max \begin{cases} I(i,j-1) - gep \\ C(i,j-1) - gep - gop \end{cases}$$

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

$$D(i,j) = \max \begin{cases} D(i-3,j) - gep \\ C(i-3,j) - gep - gop \end{cases}$$

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						

Matrix C

$$C(i,j) = \max \begin{cases} I(i,j) \\ D(i,j) \\ C(i-2,j-1) + S[b_i,a_j] - \pi \\ C(i-3,j-1) + S[b_i,a_j] \\ C(i-4,j-1) + S[b_i,a_j] - \pi \end{cases}$$

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4				
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0					
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4				
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4				
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4				
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4				
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4				
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5			
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5			
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2		
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5			
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2		
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5		
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2		
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5	5		
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5		
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	-5
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	-5
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	0
0						
0						
0						
0						

Matrix C

Sequential Implementation

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞	-5	-5	-5	-4	-5	-5
∞	-5	-5	-5	-2	-4	-5
∞	-5	-5	1	9	7	5
∞	-5	-3	-3	1	-3	-4

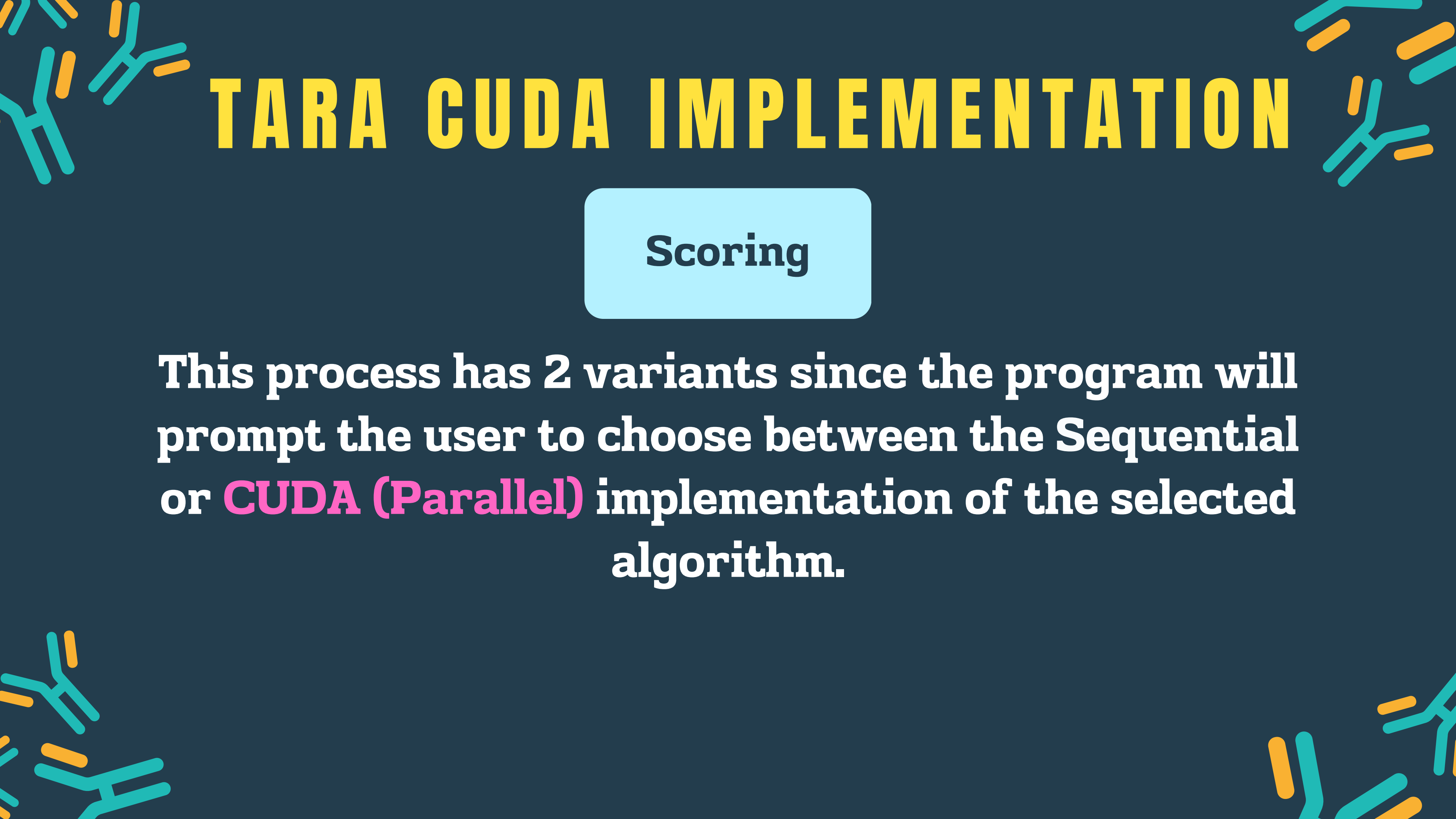
Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	-5
-5	-5	-5	-5	-5	-5	-5
-5	-5	-5	-5	-3	-5	-5
-5	-2	4	1	-3	-2	-5
-5	-5	-5	-4	-5	-5	-5

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	0
0	0	0	1	0	0	0
0	0	0	3	0	0	0
0	0	4	14	9	7	5
0	2	2	4	0	1	0

Matrix C



TARA CUDA IMPLEMENTATION

Scoring

This process has 2 variants since the program will prompt the user to choose between the Sequential or **CUDA (Parallel) implementation of the selected algorithm.**



CUDA (PARALLEL) IMPLEMENTATION

- This process includes **configuring the CUDA environment**, as well as **parallelizing the alignment scoring**.
 - Due to block synchronization limitations, we developed **a hybrid multi-level anti-diagonal wavefront implementation** using a combination of host-side loops and in-kernel block synchronization.
- 

CUDA (PARALLEL) IMPLEMENTATION

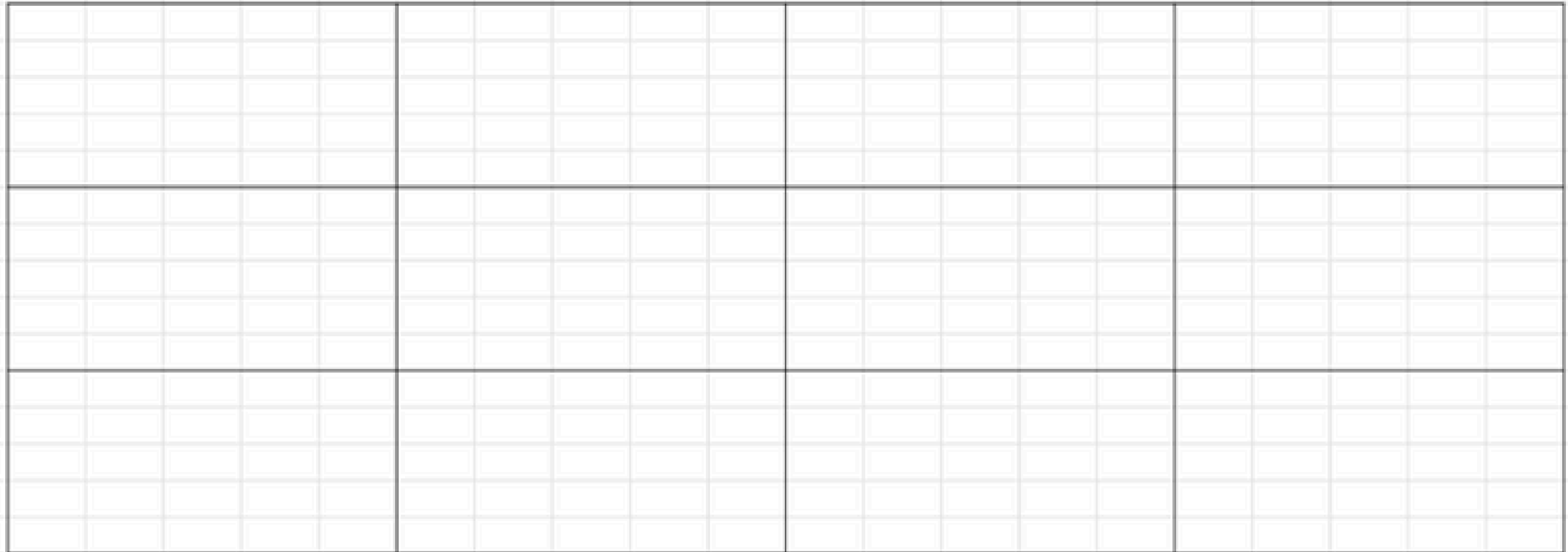


Figure 5-18: Sample Visualization of Submatrix Division

CUDA (PARALLEL) IMPLEMENTATION

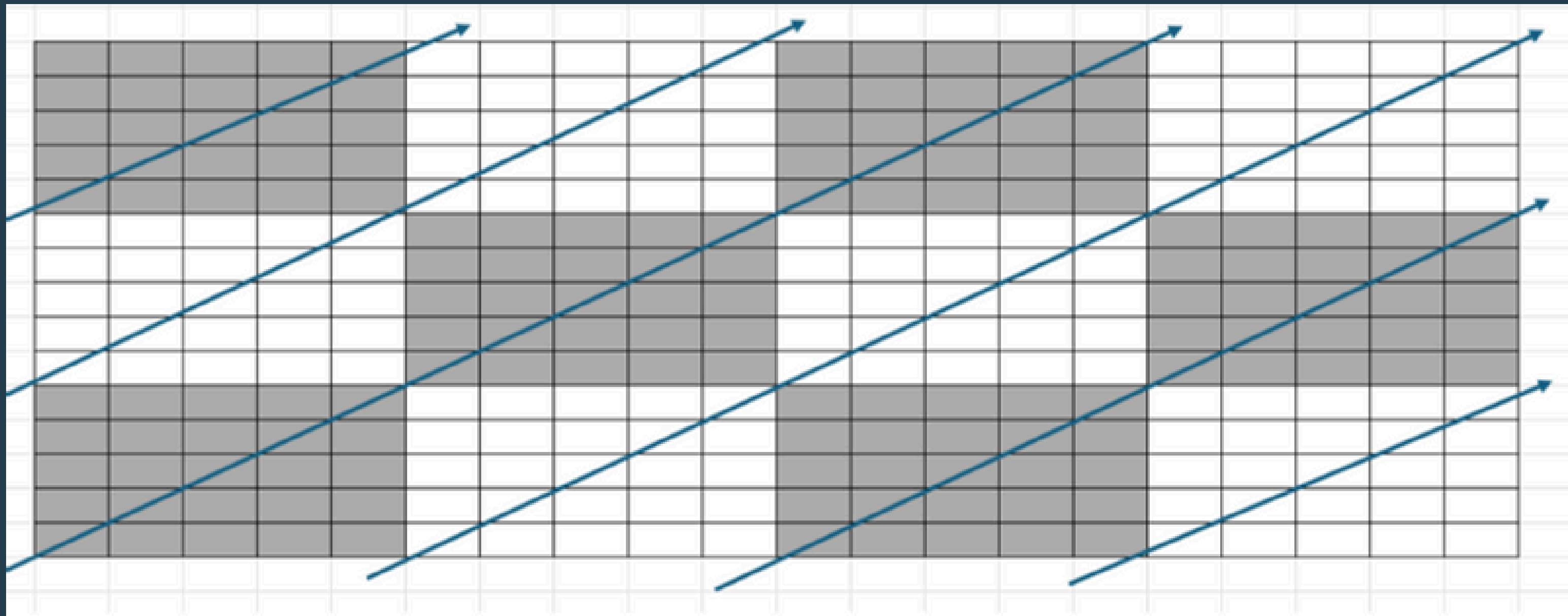
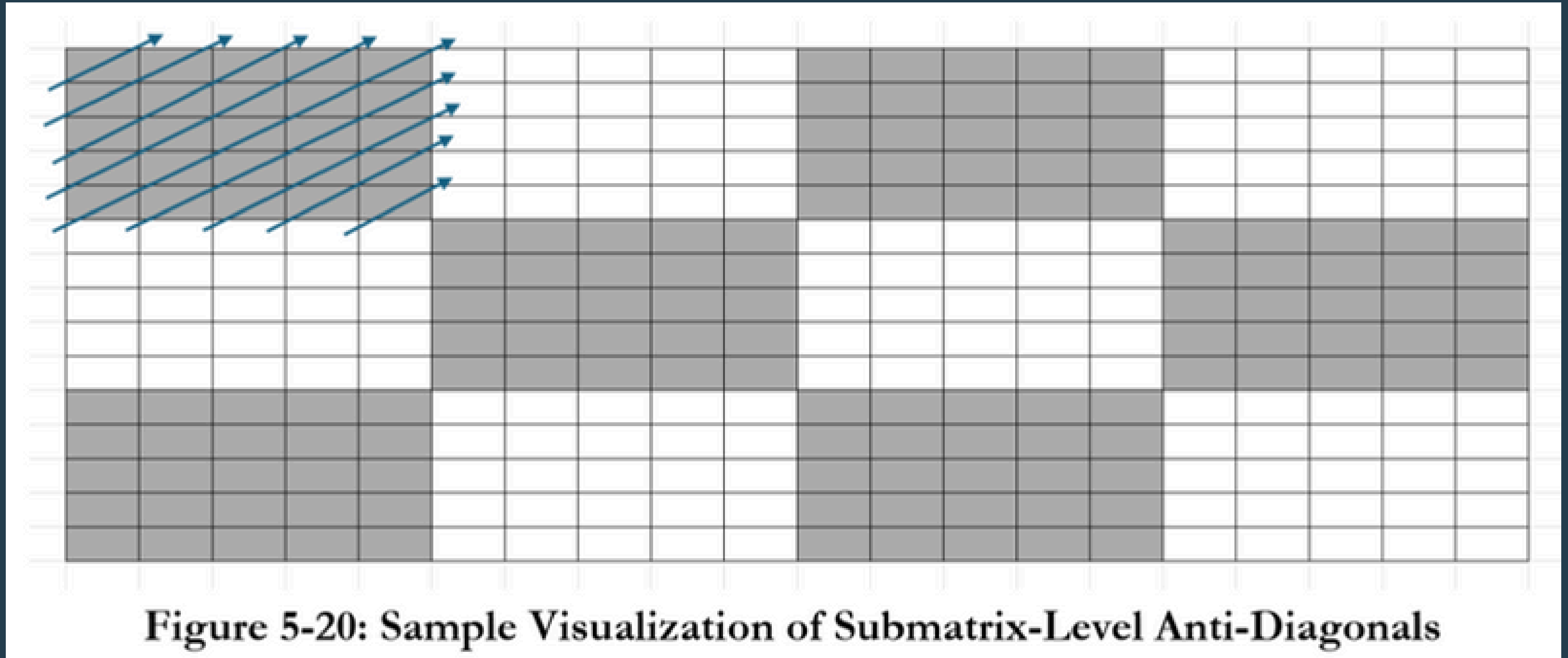


Figure 5-19: Sample Visualization of Macro-Level Matrix Anti-Diagonals

CUDA (PARALLEL) IMPLEMENTATION



CUDA (PARALLEL) IMPLEMENTATION

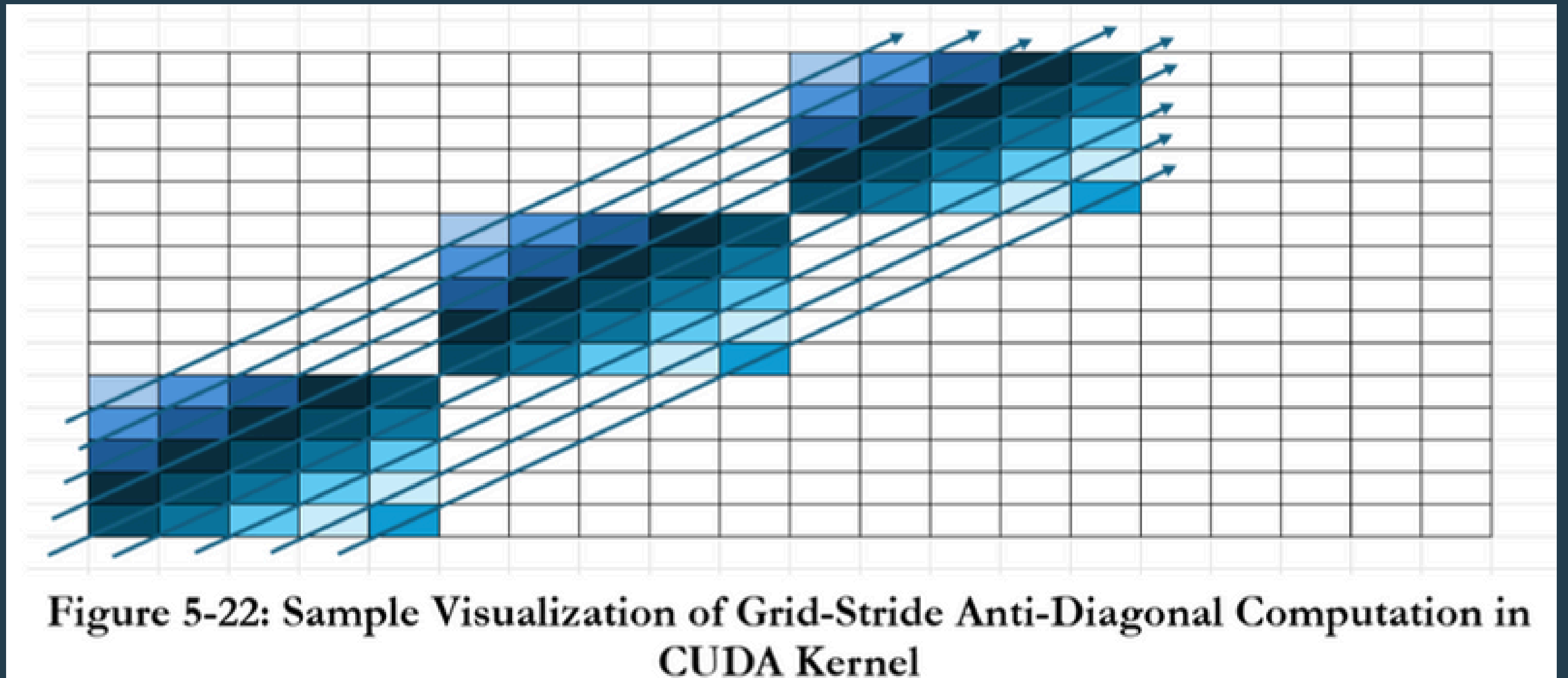
- **Host Side (CPU):** responsible for setting up data, grid and block configuration, memory management, and **launching the CUDA kernels**.
- **Block dim** is set to **(32, 32)**, **Grid dim** is adjusted to the current anti-diagonal.
- Handles **memory allocation** and **data transfers** (via *cudaMalloc* and *cudaMemcpy*).
- Manages kernel execution flow, and synchronizes the kernel launches along each anti-diagonal to **wait before the next anti-diagonal is launched**.



CUDA (PARALLEL) IMPLEMENTATION

- **Device Side (GPU):** responsible for the actual parallel computation of the scoring process using **submatrix anti-diagonal parallelization**.
 - A Block is configured **to handle a submatrix** and **strides across all submatrices** in the current anti-diagonal.
 - Threads within a block compute **all** of the **same index locations** i, j of the IDC matrices.
 - Controlled via an **anti-diagonal loop** and a **conditional** for when the thread is in the current wavefront.
- 

CUDA (PARALLEL) IMPLEMENTATION



Anti Diagonal 1

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5						
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

Anti Diagonal 1

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

$$I(i,j) = \max \left\{ \begin{array}{l} I(i,j-1) - gep \\ C(i,j-1) - gep - gop \end{array} \right.$$

Anti Diagonal 1

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

Anti Diagonal 1

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0						
0						
0						
0						
0						

Matrix C

$$D(i,j) = \max \begin{cases} D(i-3,j) - gep \\ C(i-3,j) - gep - gop \end{cases}$$

Anti Diagonal 1

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0					
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Anti Diagonal 1

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5					
∞						
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0					
5						
5						
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						

Matrix C

$$C(i,j) = \max \begin{cases} I(i,j) \\ D(i,j) \\ C(i-2,j-1) + S[b_i,a_j] - \pi \\ C(i-3,j-1) + S[b_i,a_j] \\ C(i-4,j-1) + S[b_i,a_j] - \pi \end{cases}$$

Anti Diagonal 2

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-5				
∞	-5					
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0					
-5						
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Anti Diagonal 2

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-5				
∞	-5					
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4				
-5	-5					
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1					
0						
0						
0						
0						

Matrix C

Anti Diagonal 2

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-5				
∞	-5					
∞						
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4				
-5	-5					
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0	0					
0						
0						
0						

Matrix C

Anti Diagonal 3

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞	-5	-5				
∞	-5					
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4				
-5	-5					
-5						
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0	0					
0						
0						
0						

Matrix C

Anti Diagonal 3

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞	-5	-5				
∞	-5					
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5			
-5	-5	-5				
-5	-5					
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9				
0	0					
0						
0						
0						

Matrix C

Anti Diagonal 3

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4			
∞	-5	-5				
∞	-5					
∞						
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5			
-5	-5	-5				
-5	-5					
-5						
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0	0	0				
0	0					
0						
0						

Matrix C

Anti Diagonal 4

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2		
∞	5	5	5			
∞	5	5				
∞	5					
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5			
5	5	5				
5	5					
5						
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0	0	0				
0	0					
0						
0						

Matrix C

Anti Diagonal 4

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2		
∞	-5	-5	-5			
∞	-5	-5				
∞	-5					
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5		
-5	-5	-5	-5			
-5	-5	-5				
-5	-2					
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4			
0	0	0				
0	0					
0						
0						

Matrix C

Anti Diagonal 4

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2		
∞	-5	-5	-5			
∞	-5	-5				
∞	-5					
∞						

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5		
-5	-5	-5	-5			
-5	-5	-5				
-5	-2					
-5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0	0	0	1			
0	0	0				
0	0					
0						

Matrix C

Anti Diagonal 5

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2	0	
∞	5	5	5	4		
∞	5	5	5			
∞	5	5				
∞	5					

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5	5		
5	5	5	5			
5	5	5				
5	2					
5						

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0	0	0	1			
0	0	0				
0	0					
0						

Matrix C

Anti Diagonal 5

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2	0	
∞	5	5	5	4		
∞	5	5	5			
∞	5	5				
∞	5					

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5	5	4	
5	5	5	5	5		
5	5	5	5			
5	2	4				
5	5					

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2		
0	0	0	1			
0	0	0				
0	0					
0						

Matrix C

Anti Diagonal 5

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	
∞	-5	-5	-5	-4		
∞	-5	-5	-5			
∞	-5	-5				
∞	-5					

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	
-5	-5	-5	-5	-5		
-5	-5	-5	-5			
-5	-2	4				
-5	-5					

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0	0	0	1	0		
0	0	0	3			
0	0	4				
0	2					

Matrix C

Anti Diagonal 6

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2	0	2
∞	5	5	5	4	5	
∞	5	5	5	2		
∞	5	5	1			
∞	5	3				

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5	5	4	
5	5	5	5	5		
5	5	5	5			
5	2	4				
5	5					

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0	0	0	1	0		
0	0	0	3			
0	0	4				
0	2					

Matrix C

Anti Diagonal 6

∞	5	5	5	5	5	5
∞	5	0	2	4	5	4
∞	5	5	5	5	5	5
∞	5	5	5	5	3	5
∞	5	4	4	2	0	2
∞	5	5	5	4	5	
∞	5	5	5	2		
∞	5	5	1			
∞	5	3				

Matrix I

∞	∞	∞	∞	∞	∞	∞
5	5	5	5	5	5	5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
5	0	4	5	5	4	5
5	5	5	5	5	5	
5	5	5	5	3		
5	2	4	1			
5	5	5				

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	
0	0	0	1	0		
0	0	0	3			
0	0	4				
0	2					

Matrix C

Anti Diagonal 6

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞	-5	-5	-5	-4	-5	
∞	-5	-5	-5	-2		
∞	-5	-5	1			
∞	-5	-3				

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	-5
-5	-5	-5	-5	-5	-5	
-5	-5	-5	-5	-3		
-5	-2	4	1			
-5	-5	-5				

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	0
0	0	0	1	0	0	
0	0	0	3	0		
0	0	4	14			
0	2	2				

Matrix C

Anti Diagonal 10

∞	-5	-5	-5	-5	-5	-5
∞	-5	0	-2	-4	-5	-4
∞	-5	-5	-5	-5	-5	-5
∞	-5	-5	-5	-5	-3	-5
∞	-5	-4	4	2	0	-2
∞	-5	-5	-5	-4	-5	-5
∞	-5	-5	-5	-2	-4	-5
∞	-5	-5	1	9	7	5
∞	-5	-3	-3	1	-3	-4

Matrix I

∞	∞	∞	∞	∞	∞	∞
-5	-5	-5	-5	-5	-5	-5
∞	∞	∞	∞	∞	∞	∞
∞	∞	∞	∞	∞	∞	∞
-5	0	-4	-5	-5	-4	-5
-5	-5	-5	-5	-5	-5	-5
-5	-5	-5	-5	-3	-5	-5
-5	-2	4	1	-3	-2	-5
-5	-5	-5	-4	-5	-5	-5

Matrix D

0	0	0	0	0	0	0
0	5	1	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	2	0	0
0	1	9	4	2	3	0
0	0	0	1	0	0	0
0	0	0	3	0	0	0
0	0	4	14	9	7	5
0	2	2	4	0	1	0

Matrix C

Scoring

Matrix I

TARA CUDA IMPLEMENTATION

Scoring

		A	C	G	T	M	I	T	D	V	Q	L	A	I	F	S	N	V	L	G	V	F	L	F	L	L	V	V	A	Y	H	Y	I	N	A	N	T	G	K	P	S	A	K	A	K	
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40				
A	0	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999
T	1	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
G	2	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999
A	3	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999	-999
T	4	-5	0	-4	-5	-5	-4	-5	-3	-5	-4	-5	-5	-5	-4	-3	-5	-4	-5	-3	-5	-3	-3	-4	-4	-5	-5	-5	-5	-4	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
C	5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
A	6	-5	-5	-5	-5	-3	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
C	7	-5	-2	4	-1	-3	-2	-5	-3	-4	-1	-4	-5	-5	-2	-2	-5	-2	-4	-3	-3	-3	-1	0	-1	-5	-5	-5	-5	-1	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
T	8	-5	-5	-5	-4	-5	-5	-5	-5	-4	-5	-5	-1	-4	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-4	-5	-5	-5	-5	-4	-4	-4	-4	-5	-5	-5	-1	-4	-5	-4	-5	-5	
A	9	-5	-5	-5	-2	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-4	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-3	3	-2	-4	-4	-5	-4	-5	-5	-5	-5	-5	-5	-5	-5	-5	
C	10	-5	-4	2	9	4	2	0	-2	-3	-3	-3	-3	-5	-4	-3	-4	-4	-4	-5	-5	-4	-3	-1	0	-1	-5	-5	-5	-3	-1	-5	-5	0	-5	-5	-5	-4	-5	-5	-5	-5	-5	-5	-5	
C	11	-5	-3	-3	-1	-5	-4	-5	-1	-5	-2	-5	-3	-4	-3	-1	-5	-4	-5	-1	-5	-1	-1	-4	-3	-5	-5	-5	-2	-3	-5	-5	-5	-5	-5	-5	-5	-3	-2	-5	-5	-5	-5	-5	-5	
G	12	-5	-5	-5	-4	2	-1	-3	-5	-5	-5	-2	-5	-5	-5	-5	-5	-2	-5	-2	-5	-5	-5	-5	-5	-5	2	1	10	5	3	1	-1	-3	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5
A	13	-5	-5	0	7	8	4	1	-1	-2	-4	-5	-2	-3	-5	-5	-5	-4	-5	-5	-5	-5	-5	-3	-1	0	-3	-5	-3	-5	-3	-1	-5	0	-2	-5	-5	-4	-4	-5	-5	-5	-5	-5	-5	
C	14	-5	-5	-5	-3	4	-1	-3	-3	-2	-4	-5	-5	-5	-5	-3	-3	-5	-5	-3	-5	-3	-3	-3	-5	-4	-5	-4	-4	3	-1	-2	-5	-5	-5	-5	2	-3	-4	-3	-5	-5	-5	-5		
G	15	-5	-5	-5	-5	0	1	1	-4	-5	-5	-4	-3	-5	-5	-5	-5	-5	-4	-4	-4	-4	-5	-5	-5	-5	0	2	8	7	5	2	1	-2	-5	-3	-5	-5	-5	-5	-3	-5	-3	-5	-3	
T	16	-5	-5	-2	5	10	5	4	1	-1	-3	-5	-4	-1	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-3	-2	-3	-3	-5	3	2	-3	0	-2	-1	-3	-5	-2	-5	-5	-5	-5	
C	17	-5	-5	-5	0	2	4	-1	-3	-4	-3	-5	-4	-5	-5	-5	-5	-3	-5	-5	-5	-5	-5	-4	-3	-5	-5	-5	-4	1	3	1	-2	2	-3	-5	0	3	-2	-4	-3	-5	-5	-5		
A	18	-5	-5	-5	-5	-2	3	2	-1	-4	-5	-5	-5	-3	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-2	0	6	5	7	4	2	0	-4	-3	-5	-3	-5	-3	-4	-3	-4	-3	-5	
A	19	-5	-4	-2	3	8	14	9	7	5	3	1	-1	-3	3	-2	-4	-1	-5	-4	-5	-4	-4	-1	-1	-4	-4	-5	-3	4	0	2	-2	0	-3	-3	-5	-4	-1	-5	-5	-5	-5	-5		
T	20	-5	-5	-5	-2	0	4	4	-1	-2	-5	-5	-1	-3	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-2	-5	-5	-5	-1	1	4	2	0	2	-3	-2	1	4	-1	-3	-3	-3		
T	21	-5	-5	-5	-5	-4	1	15	10	8	6	4	2	0	-2	-3	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-5	-4	-2	4	3	5	6	4	1	-2	-3	-4	-5	-2	-4	-4	-3	-3		
G	22	-5	-5	-4	1	6	12	14	9	7	5	3	2	5	1	0	-2	-3	-4	-5	-5	-5	-5	-3	-3	-3	-5	-3	-5	2	10	5	8	3	1	-1	-3	-4	-3	0	-5	-5	-5	-5		
G	23	-5	-4	-1	-4	-2	4	7	13	8	8	4	2	0	0	1	-4	-2	-5	-3	-5	-3	-3	-2	-2	-4	-3	-5	-5	1	-1	2	1	1	0	-1	-4	-1	2	1	-2	-5	-5	-5		

Matrix D

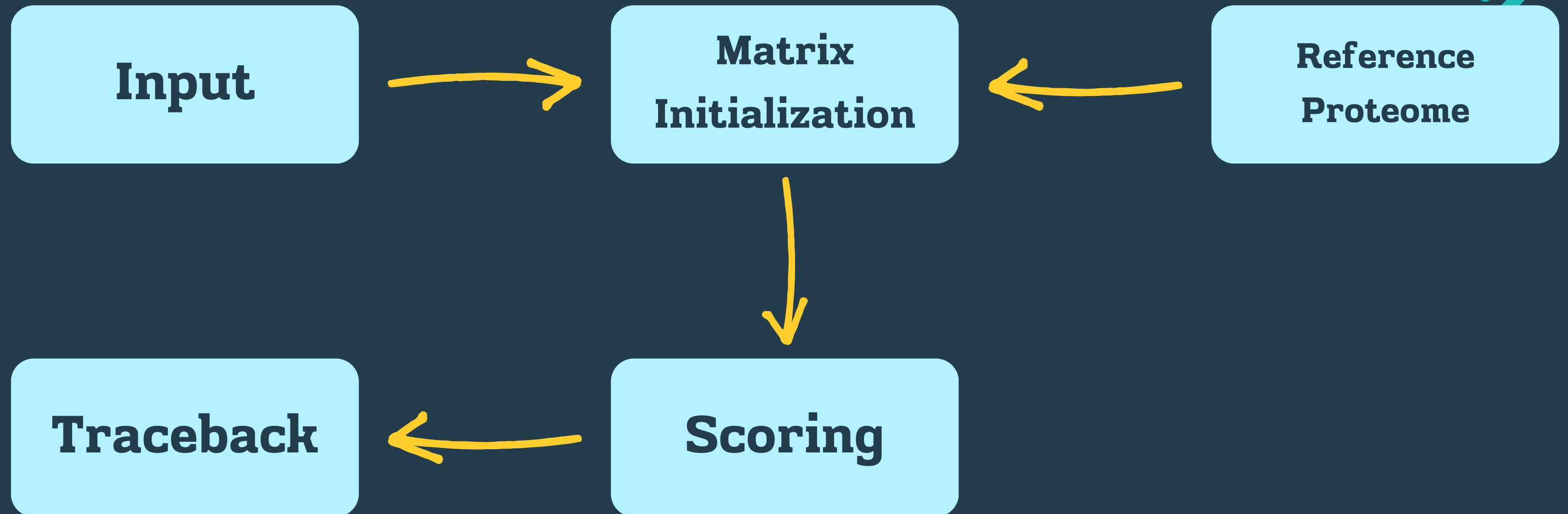
TARA CUDA IMPLEMENTATION

Scoring

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ		
1	PROTEIN HEADER [FF30]: >sp A1Z7U3 OST4_DROME Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 4 OS=Drosophila melanogaster OX=7227 GN=CG33774 PE=3 SV=1																																												
2	REF PROTEIN: MITDVQLAIFSNVLGVFLFLVVAYHYINANTGKPSAKAK																																												
3	DNA INPUT: ATGATCACTACCGACGTGAATTGGCATTCTTCCAACGTTCTGGGCGATATTTCTGCTTCCTGCTGGTCGGTCTATACATTATTAATGCCAATACTCGGAAAGCCCAGTAAGGGCAAAATGA																																												
4				M	I	T	D	V	Q	L	A	I	F	S	N	V	L	G	V	F	L	F	L	L	V	V	A	Y	H	Y	I	N	A	N	T	G	K	P	S	A	K	A	K		
5		U		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	
6	A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
7	T	1	0	5	1	0	0	1	0	2	0	1	0	0	0	1	2	0	1	0	2	0	2	2	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	G	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
9	A	3	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
10	T	4	0	1	9	4	2	3	0	2	1	4	1	0	0	3	3	0	3	1	2	2	2	4	5	4	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	C	5	0	0	0	1	0	0	0	0	1	0	0	4	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	1	1	0	0	0	4	1	0	1	0	0	
12	A	6	0	0	0	3	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	8	3	1	1	0	1	0	0	0	0	0	0	0	0	0	0	
13	C	7	0	0	4	14	9	7	5	3	2	0	2	2	0	0	2	1	0	1	0	0	1	1	4	5	4	0	0	0	0	4	0	0	5	0	0	0	1	0	0	0	0	0	
14	T	8	0	2	2	4	0	1	0	4	0	3	0	0	1	2	4	0	1	0	4	0	4	4	1	2	0	0	0	3	2	0	0	0	0	0	0	0	0	0	3	0	0	0	
15	A	9	0	0	0	0	7	4	2	0	0	0	3	0	0	0	0	0	0	3	0	3	0	0	0	0	7	4	15	10	8	6	4	2	0	0	0	0	0	0	0	0	0		
16	C	10	0	0	2	9	13	9	6	4	3	1	0	3	2	0	0	0	1	0	0	0	0	1	4	5	2	0	2	0	0	4	0	5	3	0	0	1	1	0	0	0	0		
17	C	11	0	0	0	1	9	4	2	0	3	0	0	0	0	0	0	2	0	0	0	0	0	1	2	0	1	0	1	0	8	4	3	0	0	0	0	7	2	0	2	0	0		
18	G	12	0	0	0	0	3	6	6	1	0	0	0	2	0	0	0	0	0	0	1	0	1	0	0	0	2	7	10	12	10	7	6	3	0	2	0	0	0	2	0	2			
19	A	13	0	0	0	7	15	10	9	6	4	2	0	0	4	0	0	0	0	0	0	0	0	0	0	0	2	2	2	0	8	7	2	5	0	4	2	0	3	0	0	0	0		
20	C	14	0	0	0	5	4	9	4	2	0	2	0	1	0	0	0	0	2	0	0	0	0	1	2	0	0	0	1	5	8	6	3	7	2	0	2	8	3	1	2	0	0		
21	G	15	0	0	0	0	1	8	7	4	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	8	7	12	9	7	5	1	2	0	2	0	2	1	2	0		
22	T	16	0	1	3	5	10	19	14	12	10	8	6	4	2	8	3	1	4	0	1	0	1	1	4	4	0	1	0	2	9	5	7	2	5	0	2	0	0	4	0	0	0		
23	C	17	0	0	0	1	5	9	9	4	3	0	0	4	2	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	2	6	9	7	4	7	2	0	6	9	4	2	2	0	
24	A	18	0	0	0	0	1	4	20	15	13	11	9	7	5	3	2	0	0	0	0	0	0	0	0	0	0	0	6	5	7	11	9	6	3	2	1	0	3	1	1	2	0	0	
25	A	19	0	0	0	3	8	14	19	14	12	10	8	7	10	5	5	3	0	1	0	0	0	0	0	1	2	0	2	0	4	15	10	13	8	6	4	2	1	0	5	0	0	0	
26	T	20	0	1	4	0	0	9	12	18	13	13	9	7	5	5	6	1	3	0	2	0	2	2	3	3	0	2	0	0	6	2	5	6	6	2	4	0	1	5	6	3	0	0	
27	T	21	0	2	2	0	0	5	15	24	19	17	15	13	11	9	7	5	3	1	4	0	4	4	1	1	0	0	0	4	8	5	10	8	8	3	1	0	0	1	3	0	0	0	
28	G	22	0	0	0	1	6	12	14	17	12	10	11	6	5	7	3	3	0	1	0	1	0	0	0	0	0	4	0	4	2	10	12	8	11	6	4	2	0	0	0	2	0	0	

Matrix C

TARA CUDA IMPLEMENTATION



Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	1	4
-1	-1	2	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

0	-1	-1	-2
0	0	-1	3
0	0	-1	3
-1	-1	1	4
-1	-1	2	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	2	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	2	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	3	-1	4
-1	-1	3	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	3	-1	4
-1	-1	3	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

Traceback

-1	-1	-1	-2
-1	-1	-1	3
-1	-1	-1	3
-1	-1	-1	4
-1	-1	-1	-1
3	-1	-1	4

$$TC(i, j) = \begin{cases} -2, & TC(i, j - 1) \\ -1, & TC(i - 3, j) \\ 1, & TC(i - 1, j - 1) \\ 2, & TC(i - 2, j - 1) \\ 3, & TC(i - 3, j - 1) \\ 4, & TC(i - 4, j - 1) \end{cases}$$

TARA CUDA IMPLEMENTATION

Traceback

- **Get the index of the Max Score**



Traceback

-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4



Traceback

-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

Index of the Max Score



Traceback

-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

K

TARA CUDA IMPLEMENTATION

Traceback

- **Get the index of the Max Score**
- **From here, the traceback will work backward through the matrices.**



Traceback

-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

A

K

$$TC(i,j) = \begin{cases} -2, & TC(i,j-1) \\ -1, & TC(i-3,j) \\ 1, & TC(i-1,j-1) \\ 2, & TC(i-2,j-1) \\ 3, & TC(i-3,j-1) \\ 4, & TC(i-4,j-1) \end{cases}$$



Traceback

-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

K

A

K

$$TC(i,j) = \begin{cases} -2, & TC(i,j-1) \\ -1, & TC(i-3,j) \\ 1, & TC(i-1,j-1) \\ 2, & TC(i-2,j-1) \\ 3, & TC(i-3,j-1) \\ 4, & TC(i-4,j-1) \end{cases}$$



-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

A

K

A

K

Traceback

$$TC(i,j) = \begin{cases} -2, & TC(i,j-1) \\ -1, & TC(i-3,j) \\ 1, & TC(i-1,j-1) \\ 2, & TC(i-2,j-1) \\ 3, & TC(i-3,j-1) \\ 4, & TC(i-4,j-1) \end{cases}$$



-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

SA

K

A

K

Traceback

$$TC(i,j) = \begin{cases} -2, & TC(i,j-1) \\ -1, & TC(i-3,j) \\ 1, & TC(i-1,j-1) \\ 2, & TC(i-2,j-1) \\ 3, & TC(i-3,j-1) \\ 4, & TC(i-4,j-1) \end{cases}$$



-1	-1	-1	-2	-2	-2	-2	-2
-1	-1	-1	3	-2	-2	-2	-2
-1	-1	-1	3	2	-2	-2	-2
-1	-1	-1	4	2	-2	-2	-2
-1	-1	-1	-1	3	3	-2	3
3	-1	-1	3	4	2	-2	4
-1	-1	-1	-1	-1	3	2	-2
-1	-1	-1	-1	3	3	3	-2
-1	2	-1	3	-1	3	4	2
-1	3	-1	-1	-1	2	-2	2
-1	-1	-1	-1	-1	3	-2	3
-1	-1	-1	3	-1	-1	2	4

SA

K

A

K

Traceback

$$TC(i,j) = \begin{cases} -2, & TC(i,j-1) \\ -1, & TC(i-3,j) \\ 1, & TC(i-1,j-1) \\ 2, & TC(i-2,j-1) \\ 3, & TC(i-3,j-1) \\ 4, & TC(i-4,j-1) \end{cases}$$

TARA CUDA IMPLEMENTATION

Traceback

- **Get the index of the Max Score**
- **From here, the traceback will work backward through the matrices.**
- **The traceback will continue until it encounters a cell with a score of zero**

Traceback

		M	I	T	D	V	Q	L	A	I	F	S	N	V	L	G	V	F	L	F	L	L	V	V	A	Y	H	Y	I	N	A	N	T	G	K	P	S	A	K	A	K	
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
T	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	-2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	M
G	2	0	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	
A	3	0	-2	2	2	2	-2	2	-2	-2	-2	2	2	-2	-2	2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	2	-2	2	-2	-2	-2	-2	-2	-2	-2	-2	-2	
T	4	0	3	3	-2	-2	3	-2	3	3	3	3	3	3	3	-2	3	3	3	3	3	3	3	3	3	3	3	3	3	-2	3	3	3	3	3	3	3	3	3	3	3	I
C	5	0	3	4	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	
A	6	0	3	3	2	2	3	3	3	3	2	3	3	3	3	3	3	3	3	3	3	3	2	2	2	3	-2	-2	2	3	3	3	3	3	3	3	3	3	3	3	3	
C	7	0	3	-1	3	-2	-2	-2	-2	3	3	3	3	2	3	3	3																									
T	8	0	3	3	4	-2	3	3	3	-2	3	3	-1	3	3	3	-2																									
A	9	0	3	3	-1	2	2	2	2	3	3	3	-2	3	3	3	3																									
C	10	0	3	-1	3	3	-2	-2	3	3	-2	3	3	3	3	3	3																									
C	11	0	3	3	3	4	-2	-2	-2	3	-2	3	3	3	3	-1	3																									
G	12	0	3	3	3	2	2	2	-2	-2	-2	-1	3	3	3	3	3																									
A	13	0	3	-1	-1	3	-2	3	-2	-2	-2	3	3	-2	-2	3																										
C	14	0	3	3	3	-1	3	-2	-2	-2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	2	2	2	2	2	2	-2	-2	-1	3	-2	-2	3	3	
G	15	0	3	3	3	2	2	2	3	2	-2	3	3	3	-2	3	3	3	3	3	3	3	3	3	3	-1	-1	-1	-1	3	3	3	3	-2	2	3	4	3	3	3	3	
T	16	0	3	3	-1	-1	3	-2	-2	-2	-2	-2	-2	3	-2	-2	3	-2	3	3	3	3	3	3	3	3	3	4	4	3	3	-2	3	-2	3	3	-1	2	-2	3	3	V
C	17	0	3	3	3	3	4	3	-2	3	-2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	-2	3	3	2	3	2	3	2	3	-2	-2	3	3	-2	-2	3	
A	18	0	3	3	3	2	2	3	-2	-2	-2	-2	-2	-2	2	-2	3	2	3	3	3	3	3	2	2	3	-1	-1	-1	-1	3	3	3	3	3	3	3	4	2	3	3	Q
A	19	0	3	-1	-1	-1	-1	3	-2	-2	-2	3	3	-2	3	3	-2	3	3	3	3	3	-1	3	3	3	3	3	-1	3	-2	3	-2	-2	-2	-2	3	2	2	-2	3	
T	20	0	3	3	-2	-1	4	4	2	-2	2	-2	-2	3	4	-2	3	3	3	3	3	3	3	3	3	3	-2	3	2	4	3	3	3	-1	3	-2	-1	3	3	3	-2	
T	21	0	3	3	3	-1	2	-1	3	-2	-2	-2	-2	-2	-2	-2	-2	-2	3	3	3	3	3	3	3	3	-1	-1	3	-1	2	3	2	-2	-2	-2	3	4	4	2	3	L
G	22	0	3	3	-1	-1	-1	-1	3	-2	-2	3	-2	-1	3	3	3	3	3	3	3	3	-1	-1	3	3	-2	3	-1	-1	3	-1	3	-2	-2	-2	-2	-2	-1	3	-2	
G	23	0	3	-1	3	3	-1	4	-1	2	-2	-2	-2	-2	-2	3	-2	-2	-2	-2	3	3	3	3	-2	3	3	-1	3	4	-2	4	3	-2	-2	-2	-1	3	3	-2		
C	24	0	3	3	3	2	-1	-1	3	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	3	3	3	3	-2	-2	-1	-1	3	2	3	3	3	-2	-2	3	3	3	3	-2	A	
A	25	0	3	3	-1	-1	-1	-1	4	-2	4	3	4	-2	3	-2	2	2	-2	3	3	3	3	3	3	3	-2	-2	-1	3	3	-2	3	2	-2	3	-2	3	3	2		
T	26	0	3	3	-2	3	4	-1	4	-1	3	-2	-2	-2	-2	-2	3	-2	-2	-2	-2	3	2	3	3	-2	3	3	-1	-1	3	3	-1	3	-2	3	3	-1	3	3	I	
T	27	0	3	3	3	3	2	-1	-1	-1	3	-2	-2	-2	-2	-2	-2	3	-2	3	-2	-2	3	-1	3	-2	2	-2	-2	-1	3	2	3	3	-2	3	-1	3	-1	3		
T	28	0	3	3	2	-1	-1	-1	3	-1	4	2	-2	-2	-2	-2	-2	2	-2	-2	-2	-2	-2	-2	-2	-2	4	-1	3	-2	-2	-2	-1	3	-2	3	3	-2	3	3	4	
T	29	0	3	3	3	-2	-1	-1	-1	-1	-1	3	-2	-2	-2	-2	-2	3	-2	-2	-2	-2	-2	-2	-2	3	2	4	2	-2	-1	-1	4	-1	-1	3	3	3	3	-1	3	P

Ref_Protein

MITDVQLAIFSNVLGVFLFLLVVAYHYINANTGKPSAKAK

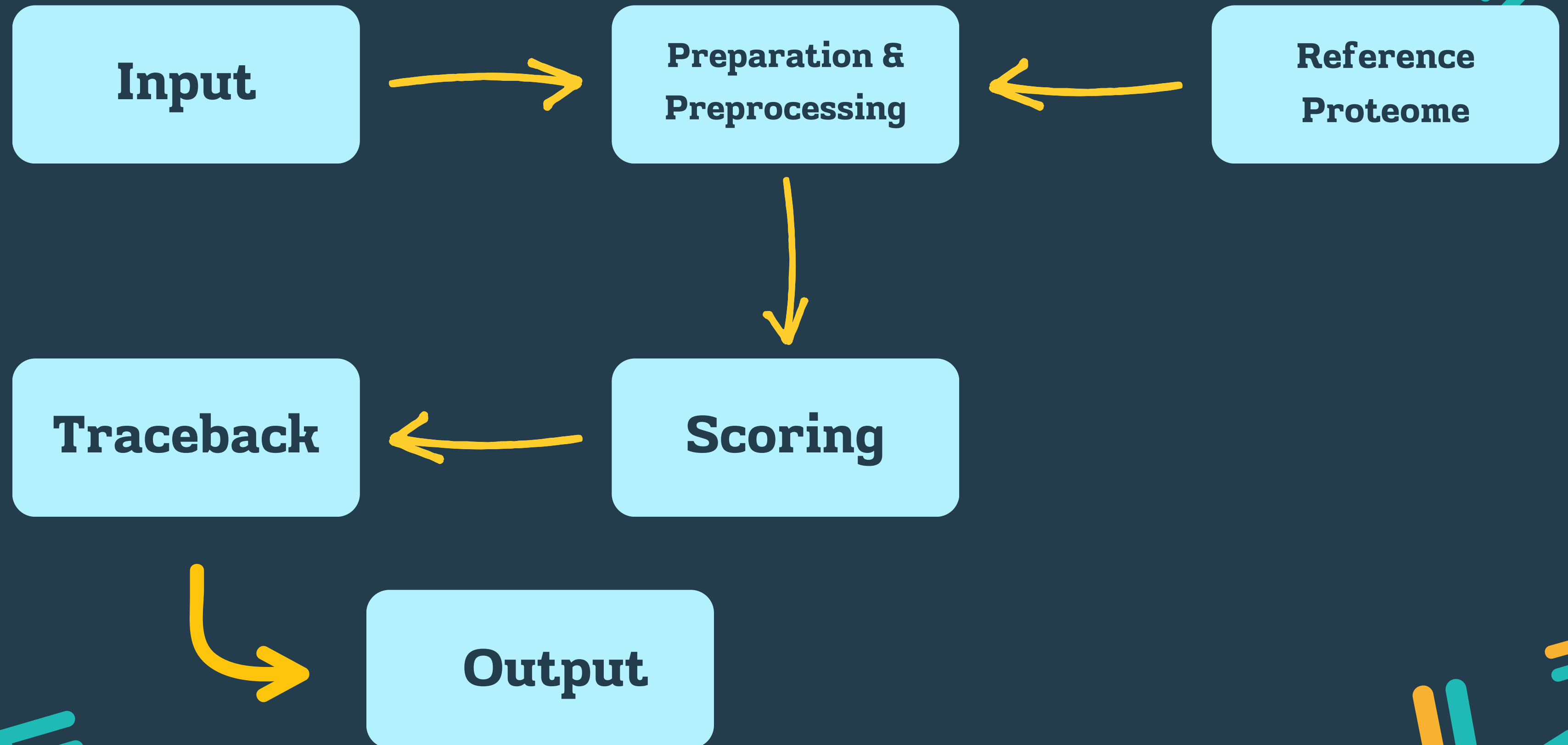
Manual Traceback

MIT-DVQLAIFSNVLGIFLFLLVGLYHYINANTGKPSAKAK

Program Traceback

MIT-DVQLAIFSNVLGIFLFLLVGLYHYINANTGKPSAKAK

TARA CUDA IMPLEMENTATION



TARA CUDA IMPLEMENTATION

REF P. MITDVQLAIFSNVLGVFL...

O. DNA MITTDVQLAIFSNVLGIFL...

O. PROT MIT-DVQLAIFSNVLGIFL...

Output

TARA CUDA IMPLEMENTATION

```
frame 1:  M I T T D V N W H F L Q R S G R Y F C F L L V G L Y I I L M P I L G K P S K A K X
frame 2:  X S L P T S I G I F S N V L G D I S A S C W S V Y T L Y X C Q Y S E S P V R Q N
frame 3:  D H Y R R Q L A F S P T F W A I F L L P A G R S I H Y I N A N T R K A Q X G K M
Output DNA: M I T T D V Q L A I F S N V L G I F L F L L V G L Y H Y I N A N T G K P S - K A K
Frameshift:      *      *      *      *      *
Output Prot:  M I T - D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A K
Reference Seq: MITDVQLAIFSNVLGVFLFLLWAYHYINANTGKPSAKAK
Time in ms: 332.675
```



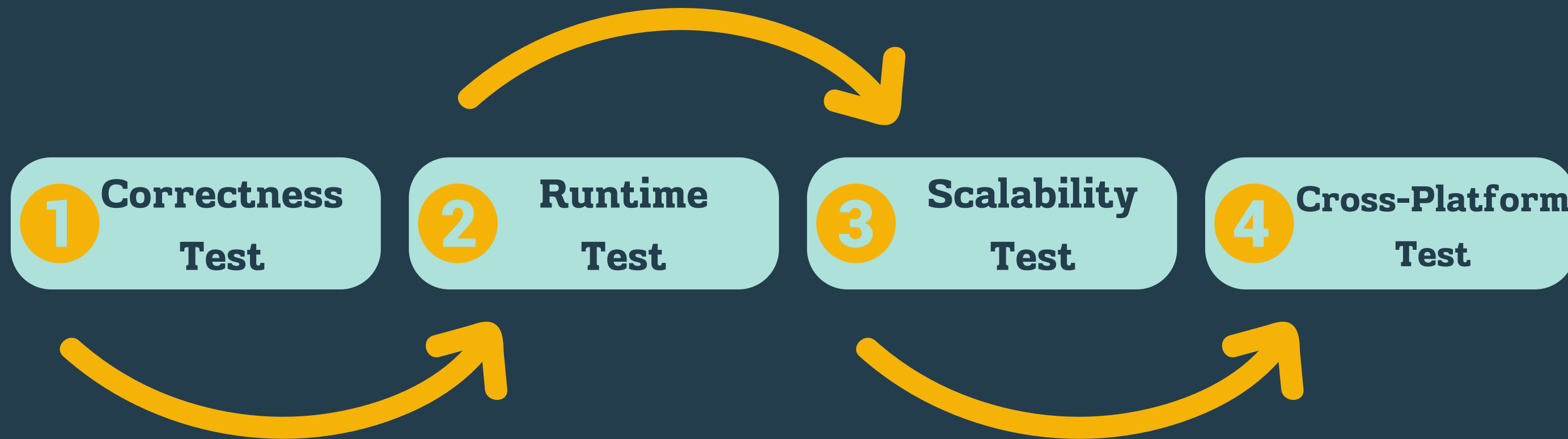
Program Output



RESULTS & ANALYSIS

RESULTS & ANALYSIS

Upon completion of development, the implementation will be subjected to a series of tests. These evaluations will focus on ensuring **correctness**, assessing **runtime performance**, verifying **scalability**, and **analyzing cross-platform performance**.



CORRECTNESS TEST

- Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where $\text{DNA}[i]$ aligns directly with $\text{Protein}[j]$ via Manual Checking and External Cross-Checking.

CORRECTNESS TEST

- Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where $\text{DNA}[i]$ aligns directly with $\text{Protein}[j]$ via Manual Checking and External Cross-Checking.
 - Perfect Alignment

Perfect Alignment

OS=Drosophila melanogaster OX=7227 GN=cytochrome c CDS=1-104
TGCGCTGCTGCTCAGCTCGGTGATCCCTGGTCAGCGG
nate dehydrogenase assembly factor
GGGCCGTGCCTCAACTACTATGATGGCCC
tryptophan protein 1 homolog Or
ATTGATTTTGTCCCAGCTCTTTGCTTACCGG



ATGATCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTCTCTGCT
GGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAAGGCAAAATGA

read30 from the fruit fly dataset*

CORRECTNESS TEST

- Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where DNA[i] aligns directly with Protein[j] via Manual Checking and External Cross-Checking.
 - Perfect Alignment
 - With Insertion

CORRECTNESS TEST



With Insertion

Dna Input v1

>1FFread30ins1

```
ATGATCACTCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGAGGCGTATTTCTGTTTCCTG
CTGGTGGTTGCCTATCATATATATTAATGCCAATACCGGAACAGCCCAGTGCCAAGGCAATG
AATGA
```

Dna Input v2

>2FFread30ins2

```
ATGATCACCGACGTGCAATTTAGGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTTCCT
GCTGGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAATAAGCCCAGTGCCAAGGCAAA
ATGA
```

Figure 6-2: DNA Substring - With Insertion

CORRECTNESS TEST

➤ Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where DNA[i] aligns directly with Protein[j] via Manual Checking and External Cross-Checking.

➤ Perfect Alignment

➤ With Insertion

➤ With Deletion

CORRECTNESS TEST



With Deletion

Dna [30]v1:

```
ATGATCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTCCCTGCT  
GGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAAGGCAAAATGA
```

Input: 108

>3FFread30del1

```
ATGATCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGTATTTTTCCTGCTGGTGGT  
TGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAAGGCAAAATGA
```

Dna [30]v2:

```
ATGATCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTCCCTGCT  
GGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAGGCAAAATGA
```

Input: 117

>4FFread30del2

```
ATGATACCGACGTGCAATTGGCCATTTTCTCCAACCTTCTGGGCGTATTTCTGTTCTGCTGGT  
GGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAGGCAAAATGA
```

Figure 6-3: DNA Substring - With Deletion

CORRECTNESS TEST

➤ Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where DNA[i] aligns directly with Protein[j] via Manual Checking and External Cross-Checking.

- Perfect Alignment
- With Insertion
- With Deletion
- With Insertion and Deletion

CORRECTNESS TEST

Dna [30]
ATGATCACTACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGATATTTCTGCTTC
CTGCTGGTGGTTGCCTATCATTATATTAATGCCAATACTCGGAAAGCCCAGTGCCAAGGCAA
AATGA

Input - 123
ATGATCACTACCGACGTCAATTGGCCATTTTCTCCAACGTTCTGGGCGATATTTCTGCTTCCTG
CTGGTCGGTCTATACATTATATTAATGCCAATACTCGGAAAGCCCAGTAAGGCAAAATGA

>read30_indelv2
ATGATTGTCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTTCCT
GCTGGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGTGCCAAGGCAAAAT
GA

>read30_indelv2
ATGATTGTCACCGACGTGCAATTGGCCATTTTCTCCAACGTTCTGGGCGTATTTCTGTTTCCT
GCTGGTGGTTGCCTATCATTATATTAATGCCAATACCGGAAAGCCCAGCAAGGCAAAATGA

Figure 6-4: DNA Substring - With Insertion and Deletion

➤ With Insertion and Deletion

CORRECTNESS TEST

➤ Evaluated the **algorithm's correctness** using aligned DNA and Protein sequence datasets, where DNA[i] aligns directly with Protein[j] via Manual Checking and External Cross-Checking.

- Perfect Alignment
- With Insertion
- With Deletion
- With Insertion and Deletion
- Complete Mismatch

AGAGTGTCTCTCCACCAATTCACATG
APSR1 PE=2 SV=1
CACGGCGCTTTATCTCGGTTCTCTA
=1
ETCATCGAGCTTTGCCAATGCCAA
TCCATTGCGATCCGAAGGTCT
TR3 PE=1 SV=1
ACTGTGCAGCGCTCTCAT
V=1
TCCGATCTAAGCAATTCGAAGGTG
TTTAAATGGATACAGAAATTTCG

TGTGCCGACGTAAATGTGATGATTATCAAACCAGGGCGGTGGTAATGGCGCTGATGATCTCA
GTCAAGAGTGCTTGTCAGCTTGGATGGTTCCCGGAGAGAGAGACTCAAGAATTGTTGGCTA
TCATAGACTTGATGTGGAATGGTTTCA



CORRECTNESS TEST

➤ Evaluated the algorithm's correctness using aligned DNA and Protein sequence datasets, where DNA[i] aligns directly with Protein[j] via **Manual Checking** and External Cross-Checking.

- Perfect Alignment
- With Insertion
- With Deletion
- With Insertion and Deletion
- Complete Mismatch

MANUAL CHECKING

PROTEIN HEADER [FF30]: >sp A1Z7U3 OST4_DROME Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 4 OS=Drosophila melanogaster OX=7227 GN=CG33774 PE=3 SV=1																											
REF PROTEIN: MITDVQLAIFSNVLGVFLFLVWAYHYINANTGKPSAKAK																											
DNA INPUT: ATGATCACTACCGACGTCAATTGGCATTCTTCTCCAACGTTCTGGGGCGATATTTCTGCTTCCTGCTGGTCTGCGGTCTATACATTATATTAATGCCAATACTCGGAAAGCCCAGTAAGGCCAAATGA																											
			M	I	T	D	V	Q	L	A	I	F	S	N	V	L	G	V	F	L	F	L	L	V	V	A	
	ij	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
A	0	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
T	1	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
G	2	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
A	3	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
T	4	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
C	5	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
A	6	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
C	7	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
T	8	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
A	9	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
C	10	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
C	11	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
G	12	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
A	13	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T
C	14	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	T



Compared the 3 scoring matrix (*c_mat*, *i_mat*, *d_mat*) from the program output to the manual excel

MANUAL CHECKING



Perfect Alignment

Score: 198

frame 1: M I T D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A K X
frame 2: X S P T C N W P F S P T F W A Y F C S C W W L P I I I L M P I P E S P V P R Q N
frame 3: D H R R A I G H F L Q R S G R I S V P A G G C L S L Y X C Q Y R K A Q C Q G K M
Output DNA: M I T D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A K
Frameshift:
Output Prot: M I T D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A K
Reference Seq: MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK
Protein Indexes: 1 to 40

Time in ms: 203.027

	Ref_Protein	MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK			
	Manual Traceback	MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK			
	Program Traceback	MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK			
	Score: 198				

MANUAL CHECKING



Perfect Alignment

Protein	Length	Score	s_index	e_index	Time(in sec)		
24	634	44	300	336	453.5861		
25	1697	50	692	722	1161.257		
26	1241	47	562	586	851.089	Top Score	
27	1481	42	281	321	1020.578	198	
28	519	47	249	287	357.5385	Found in	
29	1190	57	1024	1065	816.8593	30	
30	40	198	1	40	27.8574		
31	1134	34	224	252	787.6624		
32	1350	52	803	836	925.3988		
33	156	39	18	57	107.6762		
34	459	39	328	359	313.4625		
35	486	41	307	353	332.5456		
36	545	44	180	234	383.0587		



Time in ms: 212.524

MANUAL CHECKING



With Insertion

Sequence							
Protein	Length	Score	s_Index	e_index	Time (in ms)		
29	1190	48	1024	1065	877.3087		
30	40	184	1	40	31.1185		Top Score
31	1134	36	198	252	830.4659		184
32	1350	49	896	928	1007.453		Found in
33	156	32	21	51	126.094		30
34	459	35	192	227	337.4611		
35	486	44	313	353	356.6825		
36	545	39	201	228	394.7275		
37	2346	40	1646	1691	1712.253		
38	351	48	310	350	254.8981		
39	1977	46	1151	1195	1453.055		
40	163	35	79	113	119.6926		

MANUAL CHECKING



With Deletion

Score: 149

frame 1: M I N V Q L A I F S N V L V F F L L V V A Y H Y I N N T G K P S A K A X
frame 2: X S T C N W P F S P T F W Y F S C W W L P I I I L I I P E S P V P R H
frame 3: D Q R A I G H F L Q R S G I F P A G G C L S L Y X X Y R K A Q C Q G M
Output DNA: M I - N V Q L A I F S N V L - V F - F L L V V A Y H Y I N - N T G K P S A K A
Frameshift:
Output Prot: M I T D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A
Reference Seq: MITDVQLAIFSNVLGVFLFLLVVAYHYINANTGKPSAKAK
Protein Indexes: 1 to 39

Time in ms: 178.679

	Ref_Protein	MITDVQLAIFSNVLGVFLFLLVVAYHYINANTGKPSAKAK			
	Manual Traceback	MITDVQLAIFSNVLGVFLFLLVVAYHYINANTGKPSAKA			
	Program Traceback	MITDVQLAIFSNVLGVFLFLLVVAYHYINANTGKPSAKA			



MANUAL CHECKING



With Deletion

Protein	Length	Score	s_index	e_index	Time (in ms)		
27	1481	41	598	631	950.1122		
28	519	46	172	204	338.4439		Top Score
29	1190	54	1024	1065	809.6146		149
30	40	149	1	39	26.4733		Found in
31	1134	39	563	601	713.4803		30
32	1350	38	1300	1333	860.4029		
33	156	30	96	127	98.7649		
34	459	37	327	357	289.1428		
35	486	41	9	54	300.9241		
36	545	52	339	372	348.116		
37	2346	31	1792	1826	1504.074		
38	351	44	185	216	225.2745		
39	1977	47	598	632	1290.232		
40	163	30	79	114	103.3844		
41	858	48	158	185	188.8888		

MANUAL CHECKING



With Insertion and Deletion

Score: 147

frame 1: M I T T D V N W H F L Q R S G R Y F C F L L V G L Y I I L M P I L G K P S K A K X
frame 2: X S L P T S I G I F S N V L G D I S A S C W S V Y T L Y X C Q Y S E S P V R Q N
frame 3: D H Y R R Q L A F S P T F W A I F L L P A G R S I H Y I N A N T R K A Q X G K M
Output DNA: M I T T D V Q L A I F S N V L G I F L F L L V G L Y H Y I N A N T G K P S - K A K
Frameshift: * * * * *
Output Prot: M I T - D V Q L A I F S N V L G V F L F L L V V A Y H Y I N A N T G K P S A K A K
Reference Seq: MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK
Protein Indexes: 1 to 40

Time in ms: 203.175

Ref_Protein	MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK			
Manual Traceback	MIT-DVQLAIFSNVLGIFLFLLVGLYHYINANTGKPSAKAK			
Program Traceback	MIT-DVQLAIFSNVLGIFLFLLVGLYHYINANTGKPSAKAK			

MANUAL CHECKING



With Insertion and Deletion

Protein	Length	Score	s_index	e_index	Time (in ms)		
29	1190	46	955	990	811.477		Top Score
30	40	147	1	40	27.6793		147
31	1134	45	36	61	780.2066		Found in
32	1350	47	28	66	938.0438		30
33	156	33	103	117	106.9801		
34	459	42	150	191	313.0785		
35	486	41	305	336	335.1678		
36	545	44	344	387	372.5739		
37	2346	38	486	523	1678.204		
38	351	47	171	221	242.6604		
39	1978	50	1261	1292	1366.69		
40	163	31	72	109	114.8696		
41	652	49	327	357	474.3161		
42	950	40	296	328	676.9619		
43	883	45	51	83	615.9441		

An abstract graphic featuring several thick, rounded, geometric shapes in teal and orange colors. The shapes are arranged in a way that suggests movement or a stylized figure, set against a dark blue background.



Complete Mismatch

Scores: 28

```

frame 1:  D A V G H N M I A T C Y S R L N P P D V T E A L Q H Y Q R S I Q I D P R Q S E V V I D A C E L L V K
frame 2:  T L X A I T X L P L A T V G X T R P M X Q R P C S T T N V L F K L I R V S Q R L S L T P A N F W X
frame 3:  R C R P X H D C H L L Q X A E P A R C D R G P A A L P T F Y S N X S A S V R G C H X R L R T S G E
Output DNA: M V T E A - L A A L T N V L - - F K L I R V V R G C H X T P A N S G E
Frameshift:      *           *           *           *           *           *
Output Prot:  M I T D V Q L A I F S N V L G V F L F L L V V - A Y H Y I N A N T G K
Reference Seq: MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK
Protein Indexes: 1 to 34

```

	Ref_Protein	MITDVQLAIFSNVLGVFLFLLVWAYHYINANTGKPSAKAK			
	Manual Traceback	MVTEALAALTNVLFKLIRVV-GCHXTPANSKGK			
	Program Traceback	MVTEALAALTNVLFKLIRVV-GCHXTPANSKGK			

MANUAL CHECKING



Complete Mismatch

Protein	Length	Score	s_index	e_index	Time(in sec)	Time (in ms)	
19	635	46	147	191	535.6914		
20	1299	54	698	745	1087.269		Top Score
21	3321	60	1235	1276	2809.251		60
22	465	49	74	113	405.6597		Found in
23	1155	58	544	586	1022.003		21
24	634	44	344	389	555.4611		
25	1697	54	657	709	1464.229		
26	1241	49	879	933	1070.537		
27	1481	49	115	163	1280.757		
28	519	48	384	427	448.088		
29	1190	57	439	484	1028.707		
30	40	30	4	40	34.1434		
31	1134	52	449	489	977.2827		

MANUAL CHECKING

The Same Score

Protein	Length	Score	s_Index	e_index	Time (in ms)		
6519	151	46	4	42	165.1742		
6520	623	246	131	180	592.6589	Top Score	
6521	130	68	29	86	123.718	246	
6522	646	246	122	171	588.7805	Found in	
6523	702	246	122	171	667.0111	6520	
6524	891	49	389	433	898.3162	6522	
6525	633	50	184	225	566.7814	6523	
6526	524	51	238	280	475.9851		
6527	555	51	269	311	495.0546		
6528	604	56	235	278	544.8025		
6529	1526	56	729	772	1392.644		
6530	1375	56	578	621	1269.324		
6531	185	36	137	169	170.9544		
6532	137	36	13	52	161.6379		
6533	182	51	48	77	170.9909		

EXTERNAL CROSS-CHECKING: BLASTX

To further ensure correctness, the results of the implementation were compared with outputs from **BLASTX** and Zhang's Six Frame Algorithm in SIMD.

EXTERNAL CROSS-CHECKING: BLASTX

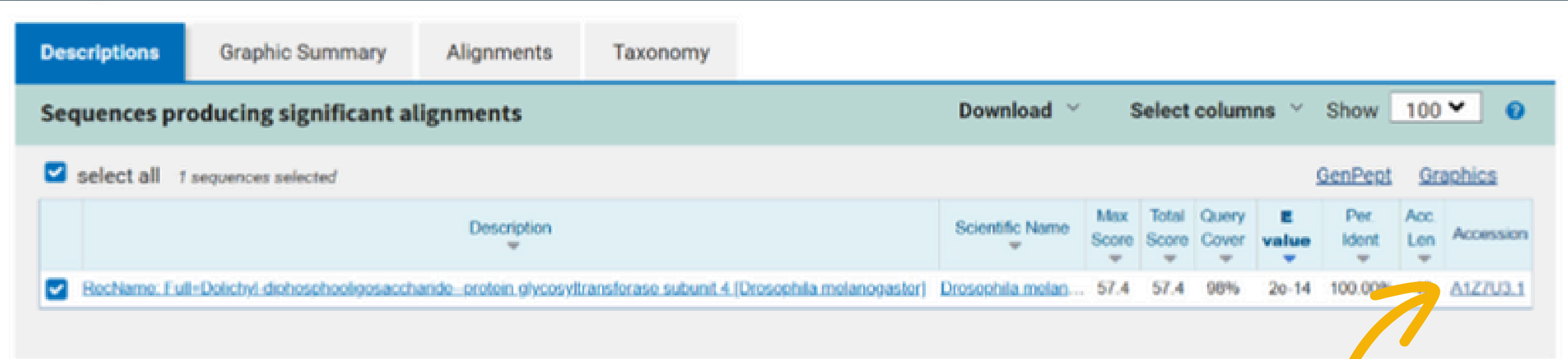


Figure 6-26: BLASTX

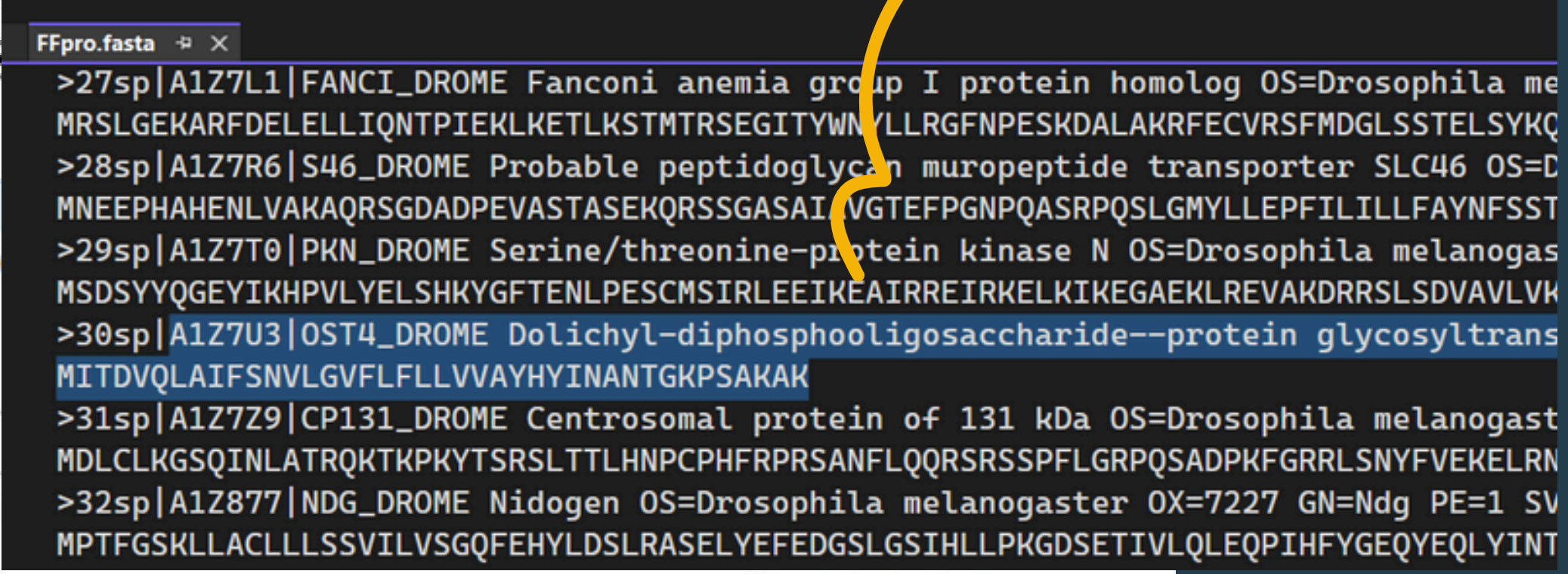
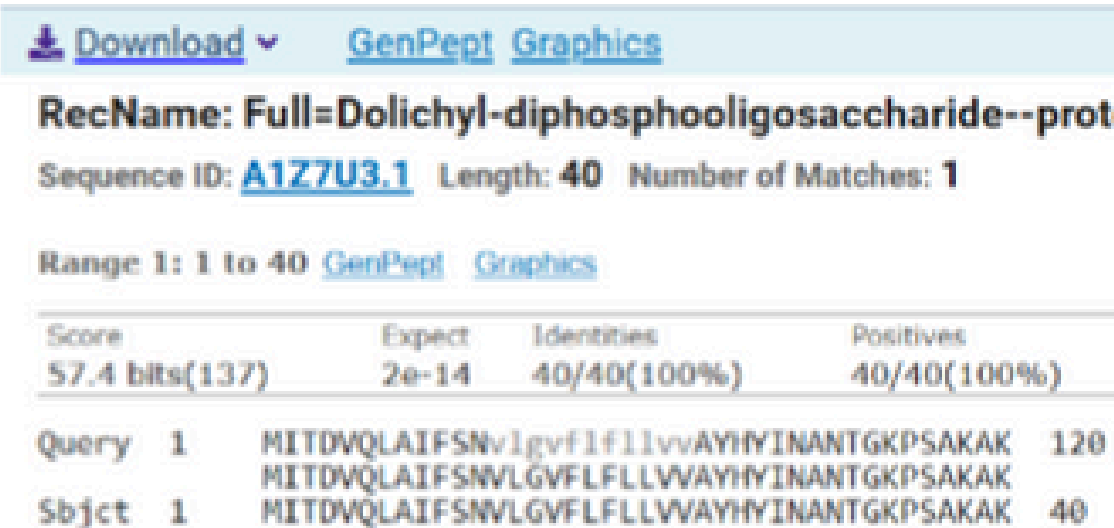
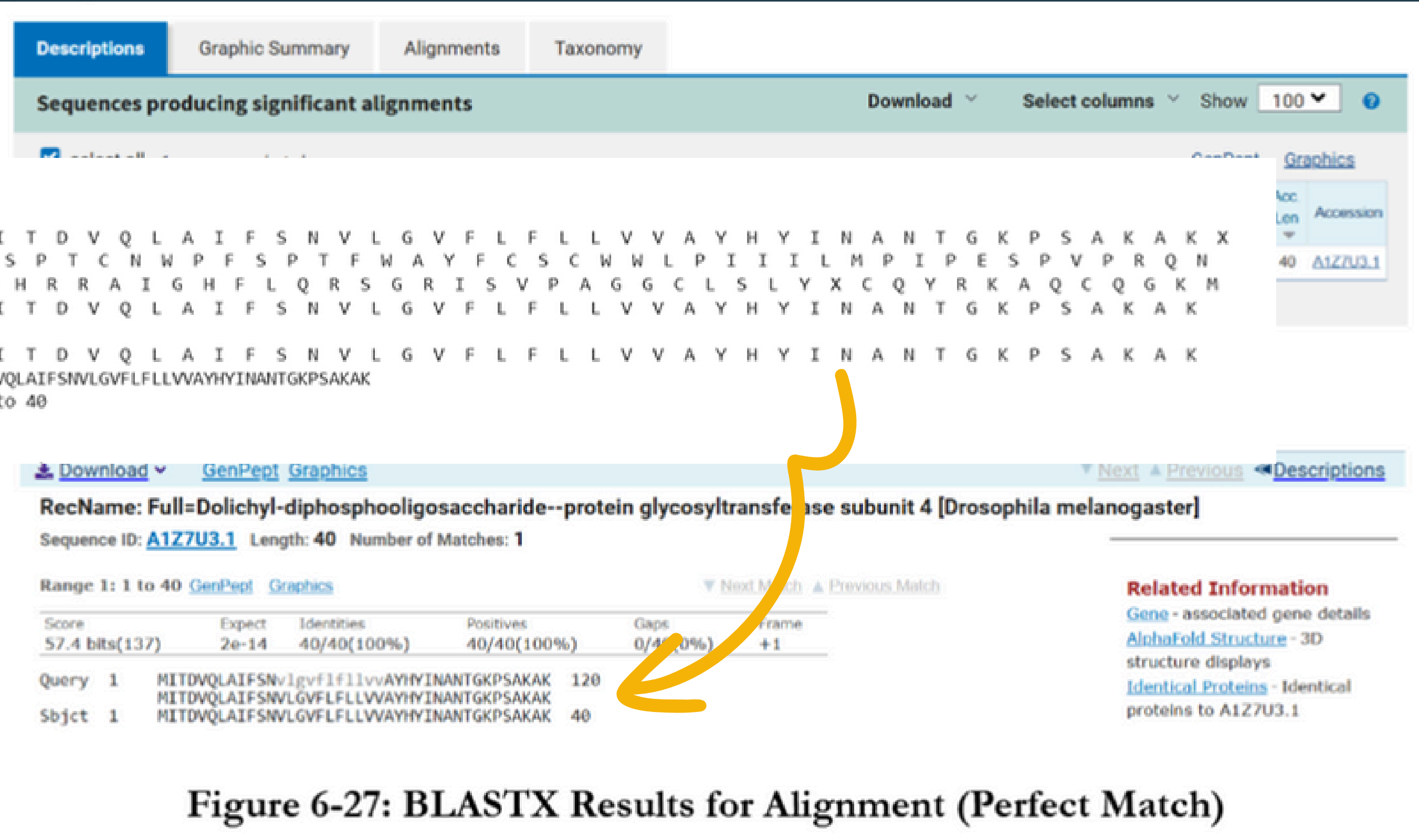


Figure 6-27: BLASTX Results for Alignment (Perfect Match)

EXTERNAL CROSS-CHECKING: BLASTX



EXTERNAL CROSS-CHECKING: BLASTX

Descriptions

Graphic Summary

Alignments

Taxonomy

Sequences producing significant alignments

Download

Select columns

Show 100

☒ select all
 1 sequences selected

[GenPept](#)
[Graphics](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	RefName: Full=Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 4 [Drosophila melanogaster]	Drosophila melan...	46.6	46.6	98%	4e-10	92.86%	40	A1Z7U3.1

Figure 6-28: BLASTX Results for With Insertion

Download
GenPept
Graphics

Next
Previous
Descriptions

RecName: Full=Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 4 [Drosophila melanogaster]

Sequence ID: A1Z7U3.1
Length: 40
Number of Matches: 1

Range 1: 1 to 40
GenPept
Graphics

Next Match
Previous Match

Score	Expect	Identities	Positives	Gaps	Frame
46.6 bits(109)	4e-10	39/42(93%)	39/42(92%)	2/42(4%)	+1

Query
1
MITDVQFRAIFSNvlgvflfllvvAYHYINANTGNKPSAKAK
126

Sbjct
1
MITDVQL-AIFSNVLGVFLFLLVVAYHYINANTG-KPSAKAK
40

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Identical Proteins - Identical proteins to A1Z7U3.1

Figure 6-29: BLASTX Results for Alignment (With Insertion)

EXTERNAL CROSS-CHECKING: BLASTX



No significant similarity found. For reasons why, [click here](#)

Figure 6-34: BLASTX Results for Complete Mismatch

EXTERNAL CROSS-CHECKING: SIMD

To further ensure correctness, the results of the implementation were compared with outputs from BLASTX and **Zhang's Six Frame Algorithm in SIMD**.

EXTERNAL CROSS-CHECKING: SIMD

Protein	Length	Score (SIMD)	Score (CUDA)		
0	774	58	58		TRUE
1	88	37	37		TRUE
2	2718	67	67		TRUE
3	1866	57	57		TRUE
4	965	61	61		TRUE
5	780	67	67		TRUE
6	343	47	47		TRUE
7	834	58	58		TRUE
8	2036	74	74		TRUE
9	400	57	57		TRUE
10	2176	245	245		TRUE
11	446	62	62		TRUE
12	668	55	55		TRUE
13	279	46	46		TRUE
14	4076	64	64		TRUE

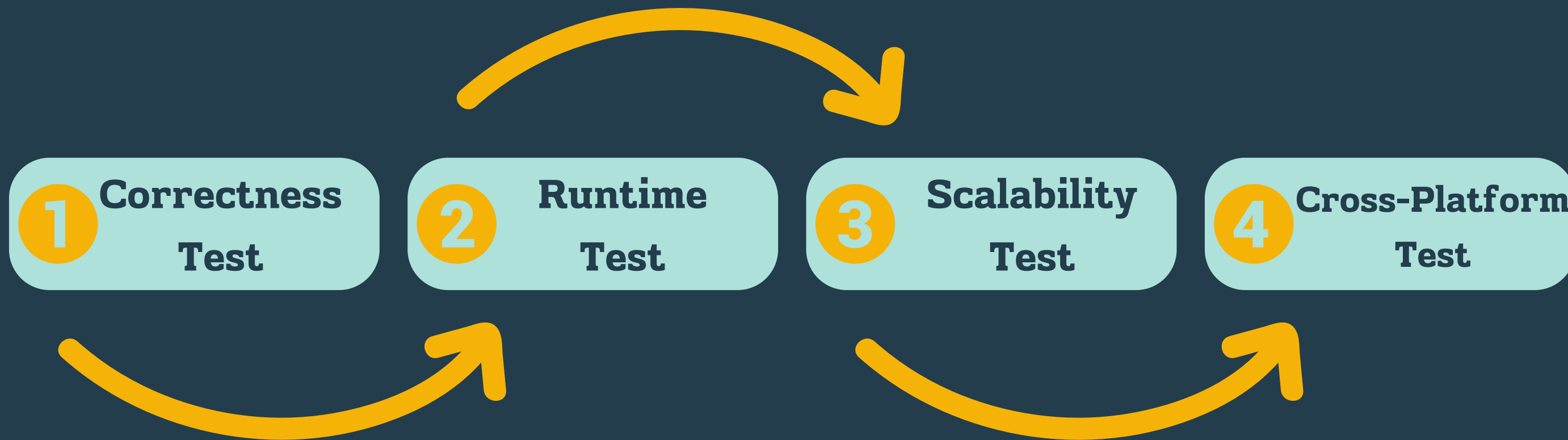
Figure 6-35: Excel Snapshot of SIMD and CUDA Score Results Comparison

CORRECTNESS TEST

	Manual Checking	BLASTX	Zhang's Six Frame in SIMD
Zhang's Six Frame in SIMT	✓	✓	✓

RESULTS & DISCUSSION

Upon completion of development, the implementation will be subjected to a series of tests. These evaluations will focus on ensuring **correctness**, assessing **runtime performance**, verifying **scalability**, and **analyzing cross-platform performance**.



RUNTIME TEST

- This test aimed to **assess the runtime performance of the CUDA implementation in comparison** to the sequential and SIMD implementations. This was done by recording the time before and after the execution of the kernel and computing the elapsed time.
- There would be two variants of this test: **Sequential vs SIMT** and **SIMD vs SIMT**.

RUNTIME TEST: SEQUENTIAL VS SIMT



PC uses

- CPU: **Intel i7-12700F** (12-core, 20-thread)
- GPU: **NVIDIA RTX 4070 Super**
- Memory: **32GB** DDR4



Visual Studio is configured to

- CUDA Runtime: shared **CUDA Runtime Library**
- Optimization: Full Optimization (**/Ox**)
- C/C++ Runtime: Multi-Threaded (**/MT**)

RUNTIME TEST: SEQUENTIAL VS SIMT

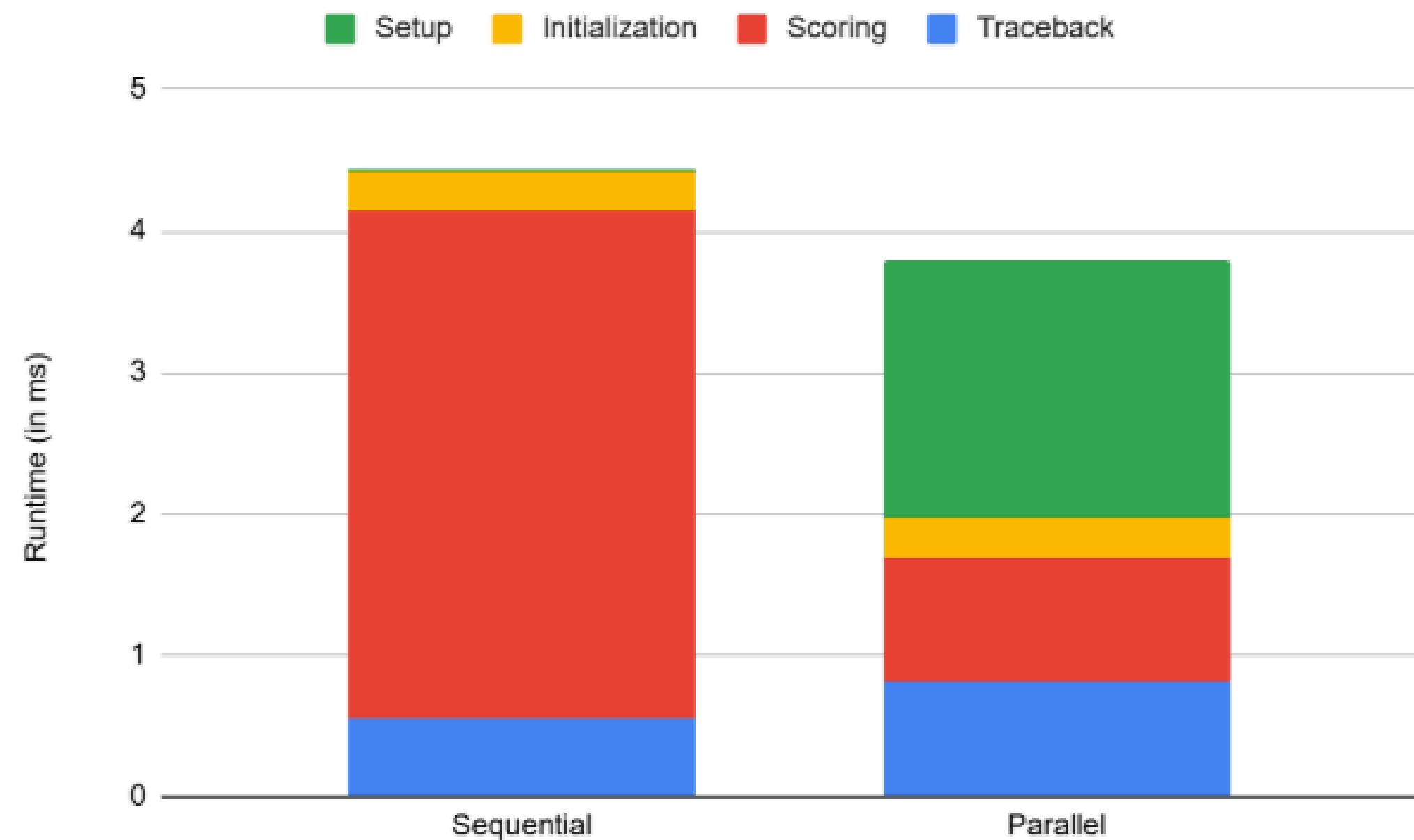


Figure 6-45: Visualization of Average Runtime Contribution of Each Stage For Single Read

- Scoring is a significant portion of the program

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

Stage	Sequential (in ms)	Fraction of the program	CUDA (in ms)	Fraction of the program
Program Setup	0.0117	0.27%	1.8238	48.09%
Matrix Initialization	0.2768	6.24%	0.2748	7.25%
Matrix Scoring	3.5892	80.98%	0.8726	23.01%
Traceback	0.5544	12.51%	0.8216	21.66%
Total	4.4321	100%	3.7928	100%

Table 6-5: Average Runtime Contribution of Each Phase

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

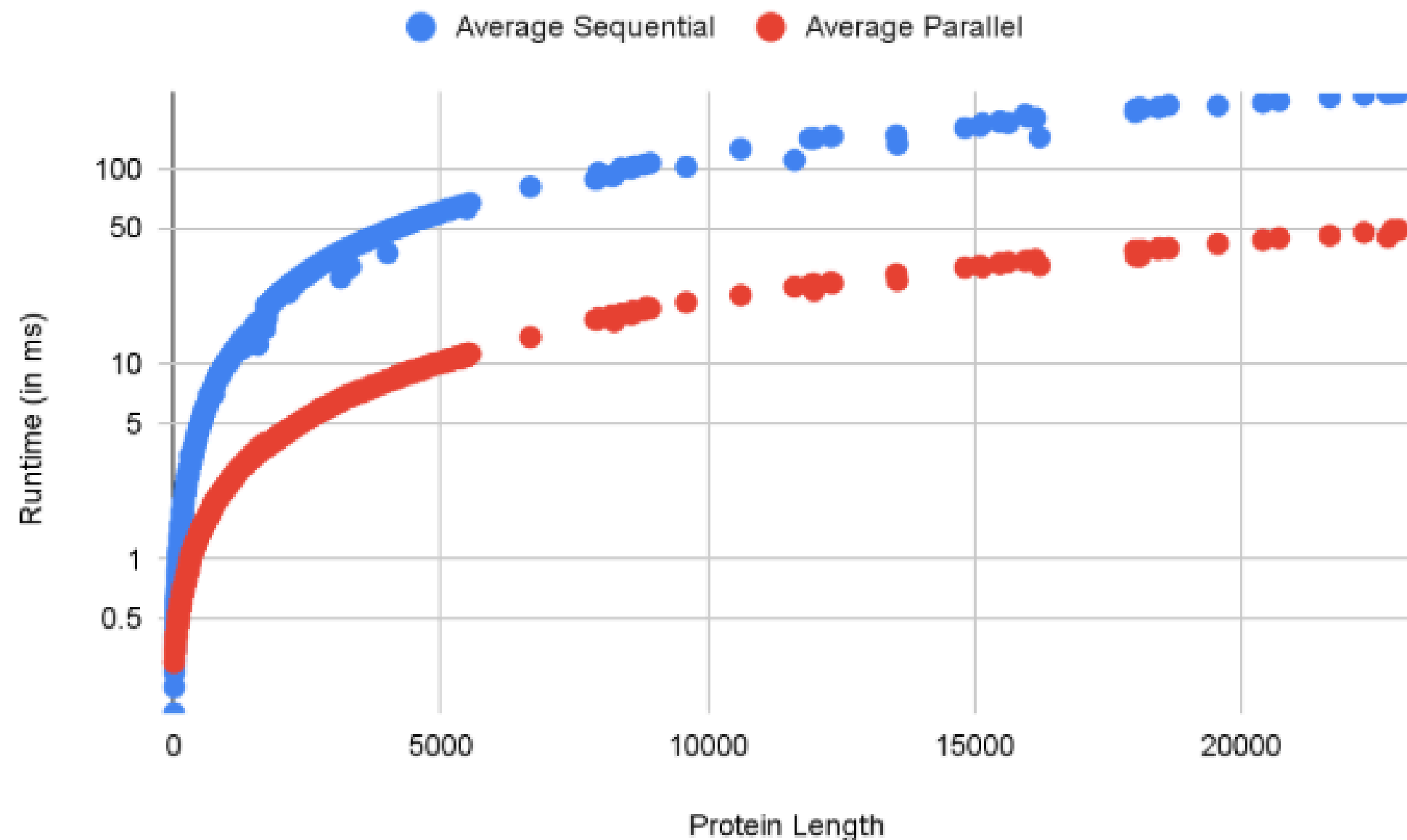


Figure 6-46: Runtime vs. Sequence Length for Sequential and Parallel Implementations (for Fruit Fly Dataset)

- CUDA performs better than seq, especially for longer reads

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

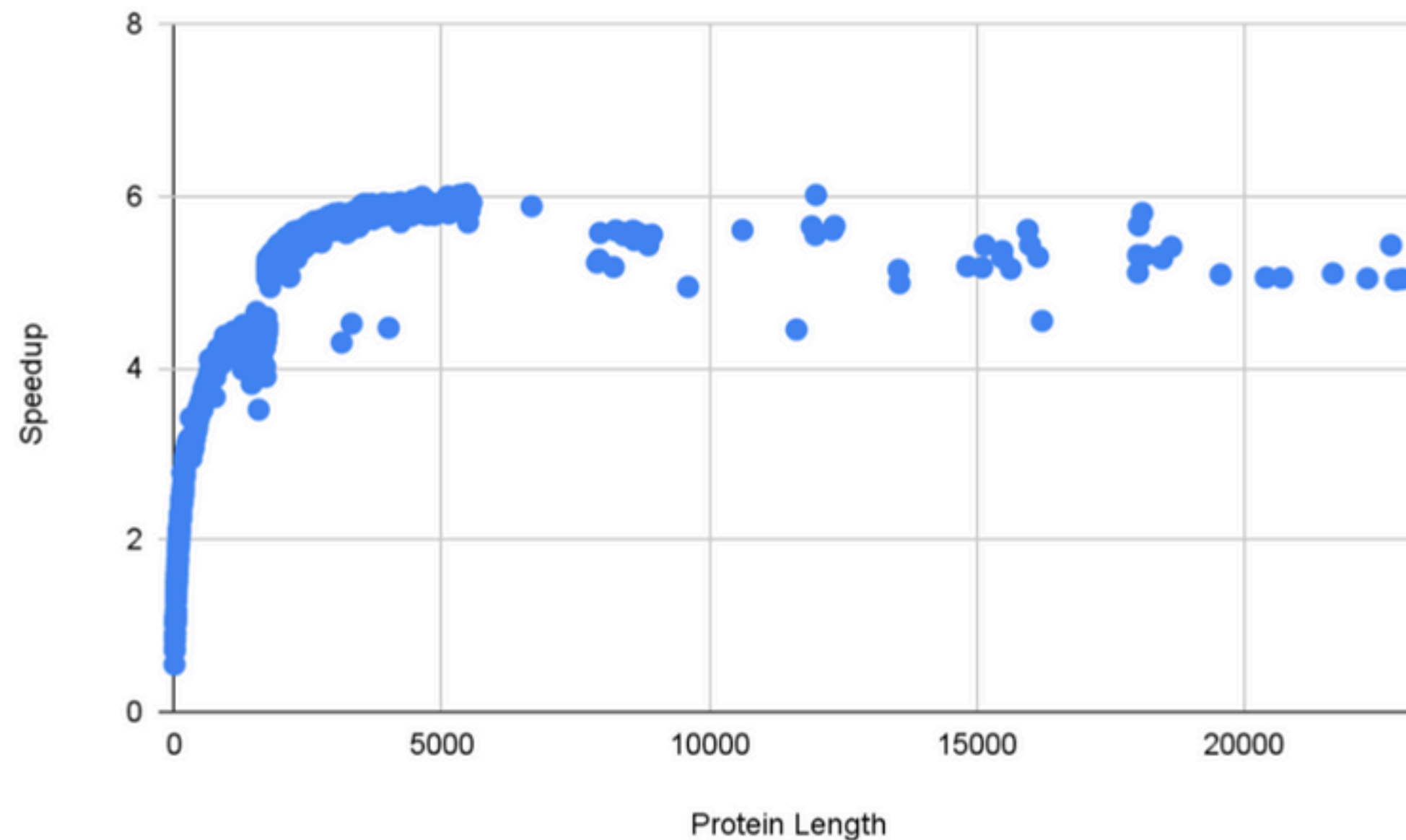


Figure 6-47: Speedup vs. Sequence Length for Sequential and Parallel Implementations (for Fruit Fly Dataset)

- Speedup generally increases as sequence length increases
- CUDA averages about a 3.51x speedup over Sequential

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

	Sequential (in ms)	SIMT (in ms)	Speedup
Average	7.1402	1.7300	3.5104x
Lowest	0.1606	0.2903	0.5468x
Highest	245.6083	48.7752	6.0252x

Table 6-6: Average Runtime and Speedup for Sequential and SIMT Implementation (for Fruit Fly Dataset)

Sequential (in s)	SIMT (in s)	Speedup
153.1505	37.1068	4.1273x

Table 6-7: Total Runtime and Speedup for Sequential and SIMT Implementation (for Fruit Fly Dataset)

- Slowdowns for shorter sequences
- Max Speedup of 6.03x
- Total Speedup of 4.13x

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

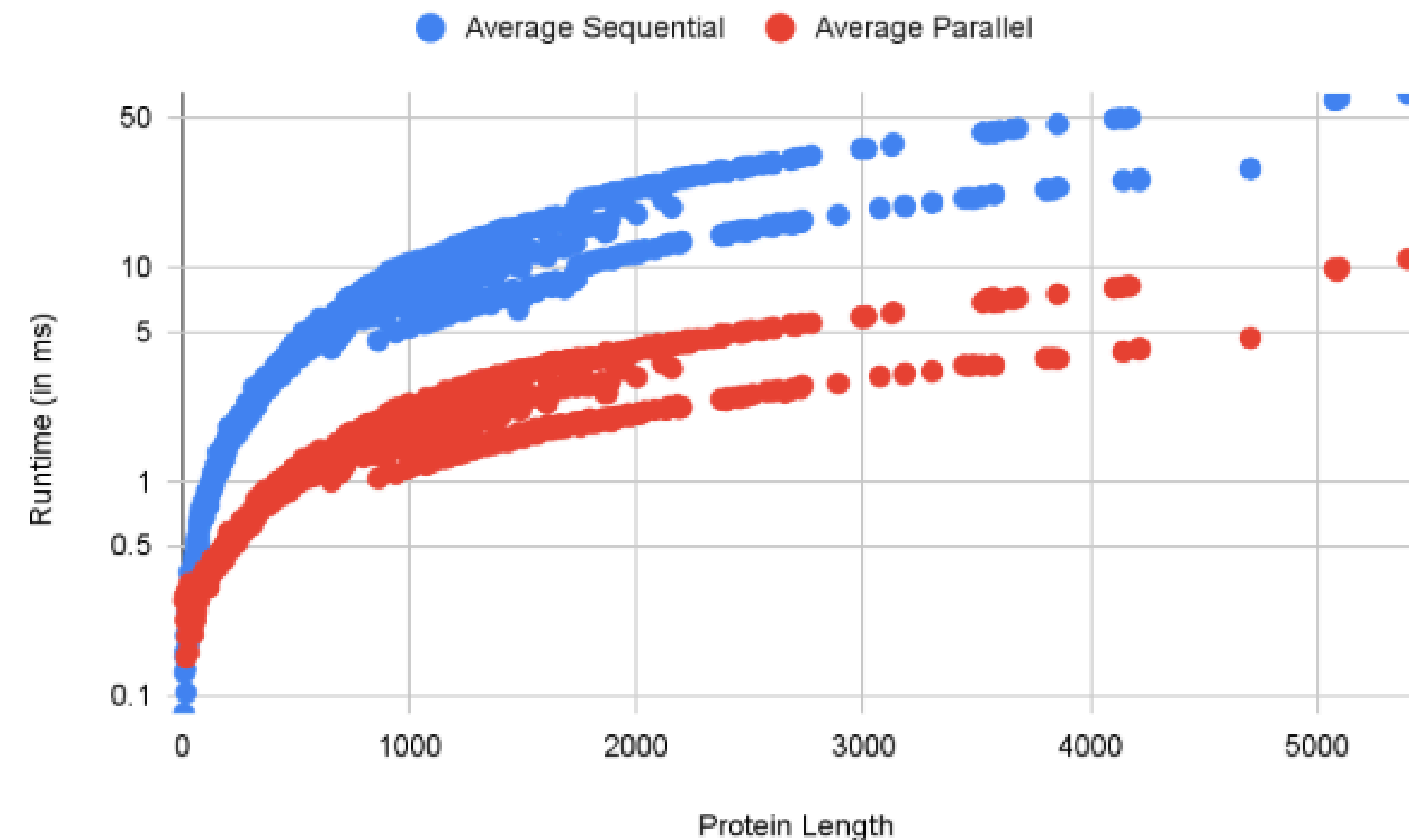
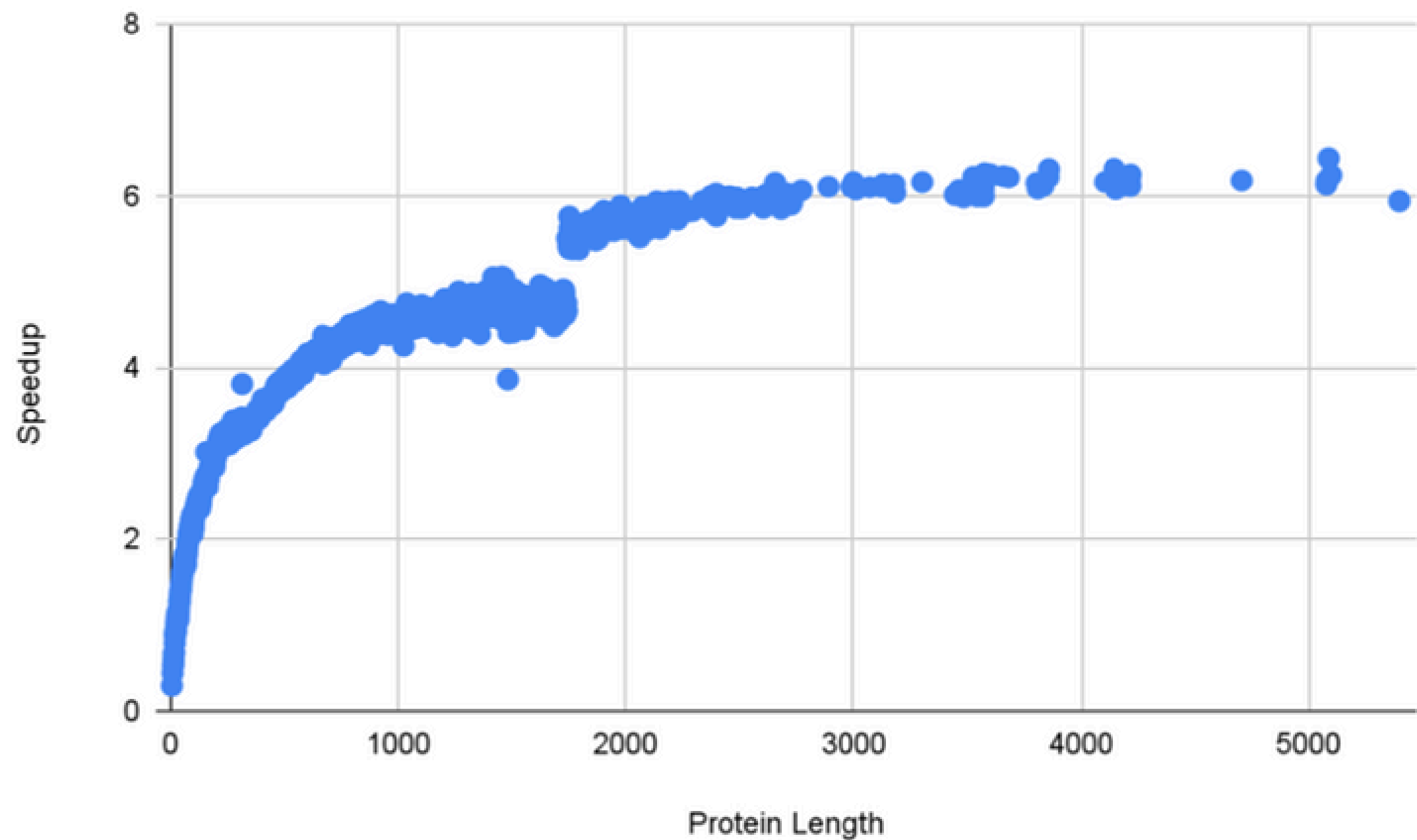


Figure 6-48: Runtime vs. Sequence Length for Sequential and Parallel Implementations (for Mouse Ear Cress Dataset)

- Behavior is consistent across datasets

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT



- Behavior is consistent across datasets
- CUDA averages about a 3.44x speedup over Sequential

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SEQUENTIAL VS SIMT

	Sequential (in ms)	SIMT (in ms)	Speedup
Average	3.4489	0.9023	3.4412x
Lowest	0.0822	0.1450	0.2954x
Highest	65.0062	10.9524	6.4371x

Table 6-8: Runtime and Speedup for SIMT and Sequential Implementation (for Mouse Ear Cress Dataset)

Sequential (in s)	SIMT (in s)	Speedup
134.9088	35.2936	3.8224x

Table 6-9: Total Runtime and Speedup for Sequential and SIMT Implementation (for Mouse Ear Cress Dataset)

- Behavior is consistent across datasets
- Max Speedup of 6.43x
- Total Speedup of 3.82x

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

RUNTIME TEST: SIMD VS SIMT

- 
- Laptop utilizes the following specs:
 - CPU: **Ryzen 7 5800H** (8-core, 16-thread)
 - GPU: **NVIDIA RTX 3050 Ti Mobile**
 - Memory: **16GB** DDR4
 -  **Compatibility issues** with the SIMD exe (Would not run with windows 11 systems).

RUNTIME TEST: SIMD VS SIMT

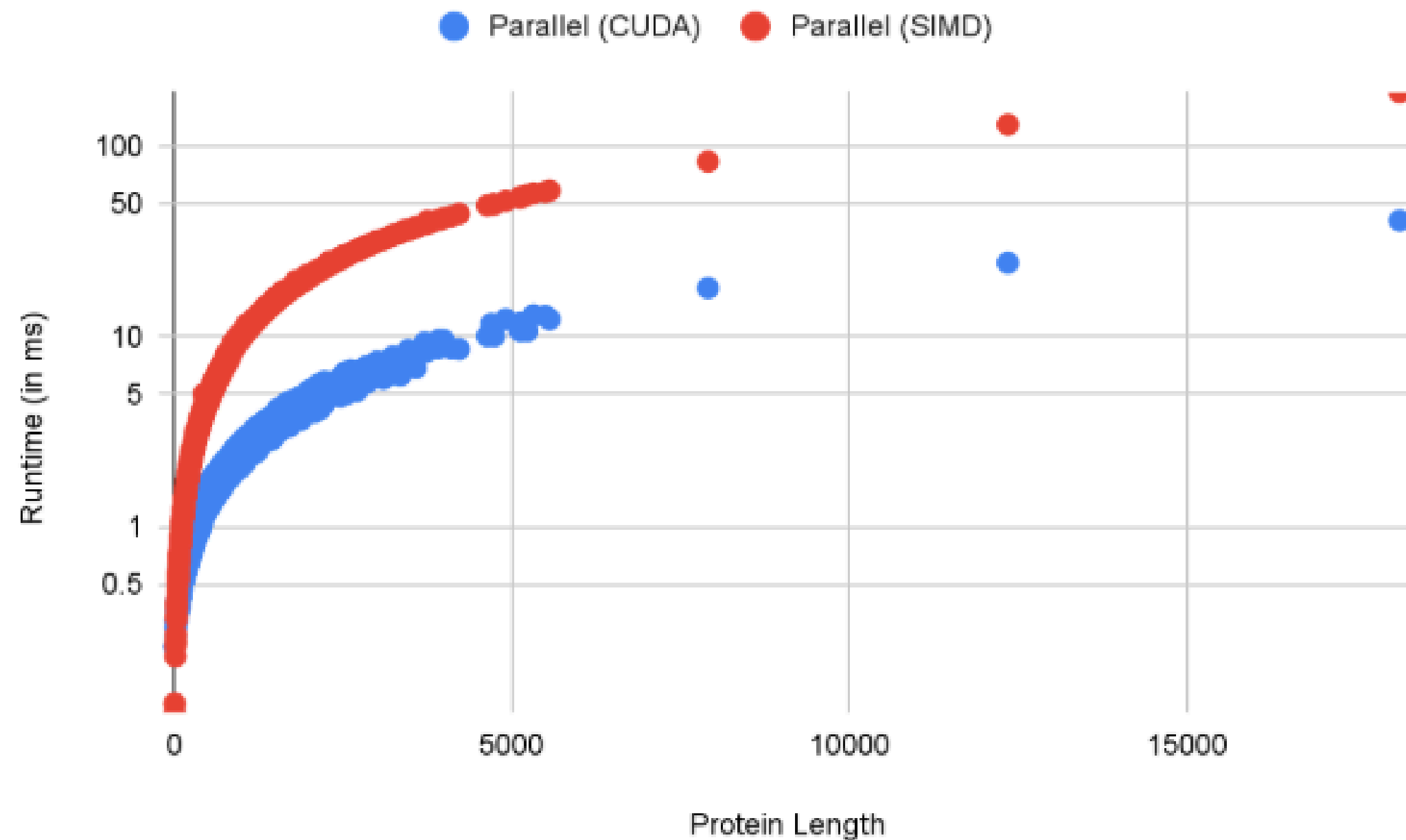


Figure 6-52: TARA CUDA Parallel Implementation vs FICS Parallel (SIMD) Implementation

- CUDA is faster than AVX2 (SIMD) by 3.14x

Tested on laptop with an NVIDIA RTX 3050 Ti Laptop GPU and Ryzen 7 5800H processor (Ox)

RUNTIME TEST: SIMD VS SIMT

	SIMD (in ms)	SIMT (in ms)	Speedup
Average	6.4093	1.7712	3.1435x
Lowest	0.1064	0.2353	0.4518x
Highest	192.3529	40.8517	5.6077x

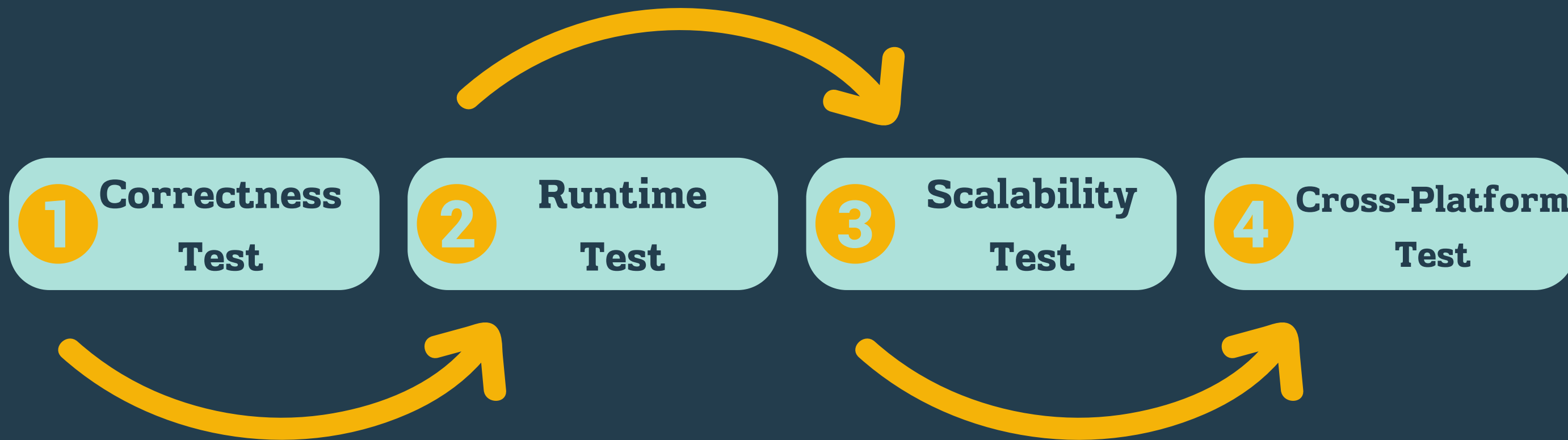
Table 6-10: Average Runtime and Speedup for SIMD and SIMT Implementation

- Slowdowns for shorter sequences
- Max Speedup of 5.61x

Tested on laptop with an NVIDIA RTX 3050 Ti Laptop GPU and Ryzen 7 5800H processor (0x)

RESULTS & DISCUSSION

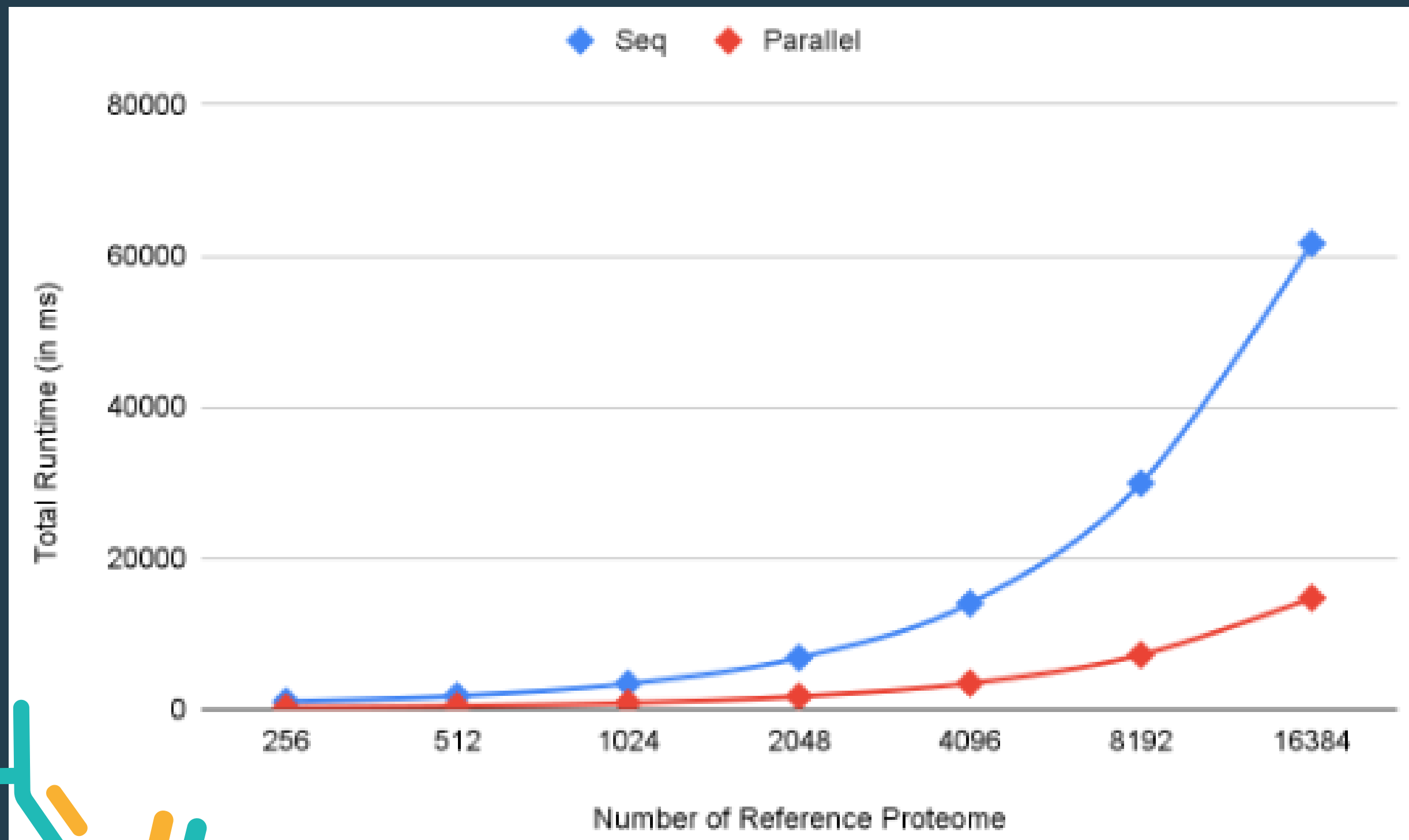
Upon completion of development, the implementation will be subjected to a series of tests. These evaluations will focus on ensuring **correctness**, assessing **runtime performance**, verifying **scalability**, and **analyzing cross-platform performance**.



SCALABILITY TEST

➤ **Two scalability tests** were conducted: input scalability and compiler optimization. **Input scalability** examines how the runtime grows with *increasing proteome input volume*. While **compiler optimization** measures the *impact of different compiler optimization levels* on execution speed.

SCALABILITY TEST: INPUT



- Polynomial growth
- Parallel implementation exhibits better scalability

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

SCALABILITY TEST: INPUT

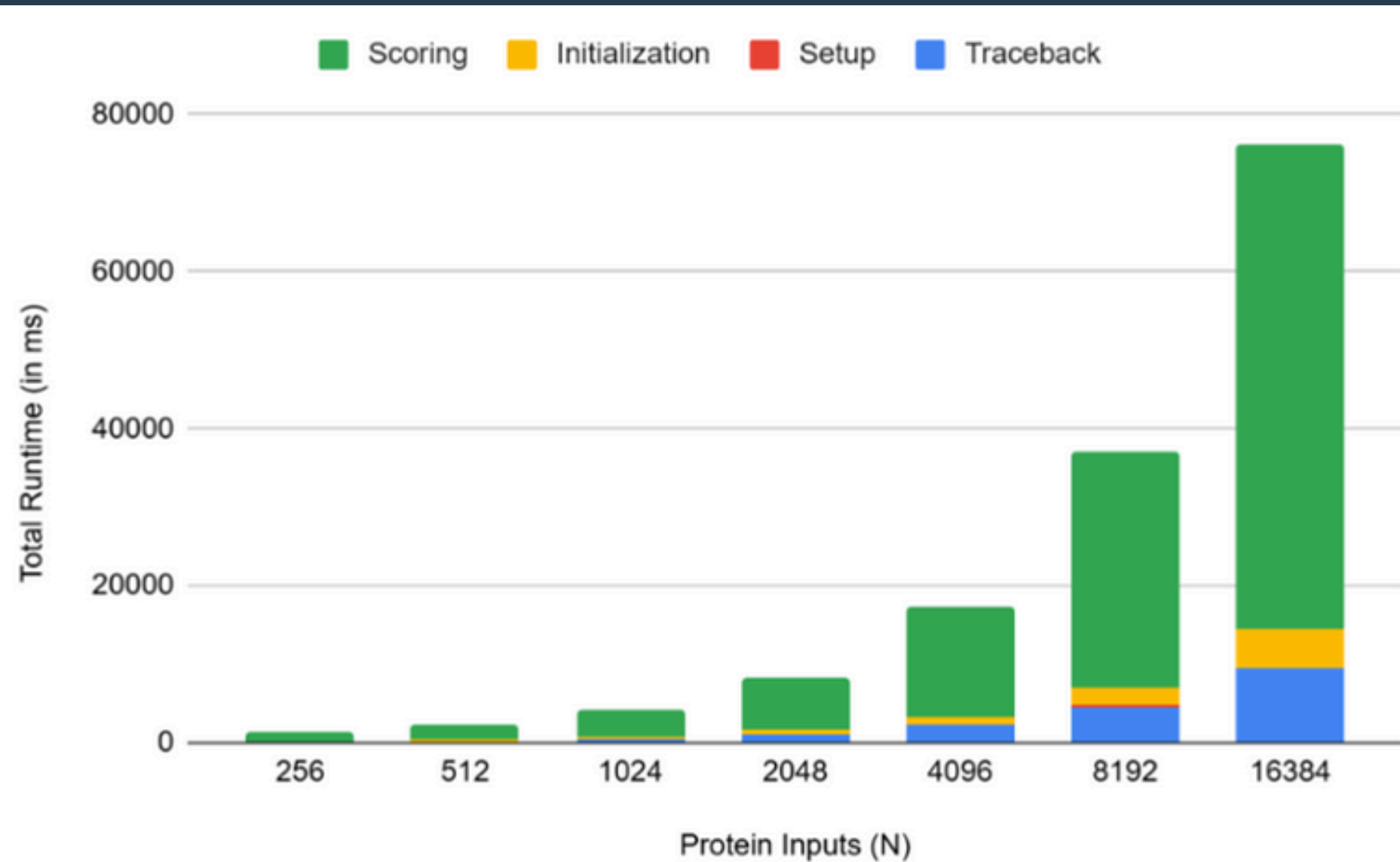


Figure 6-57: Scalability of the Runtime Contribution of Each Phase (Sequential)

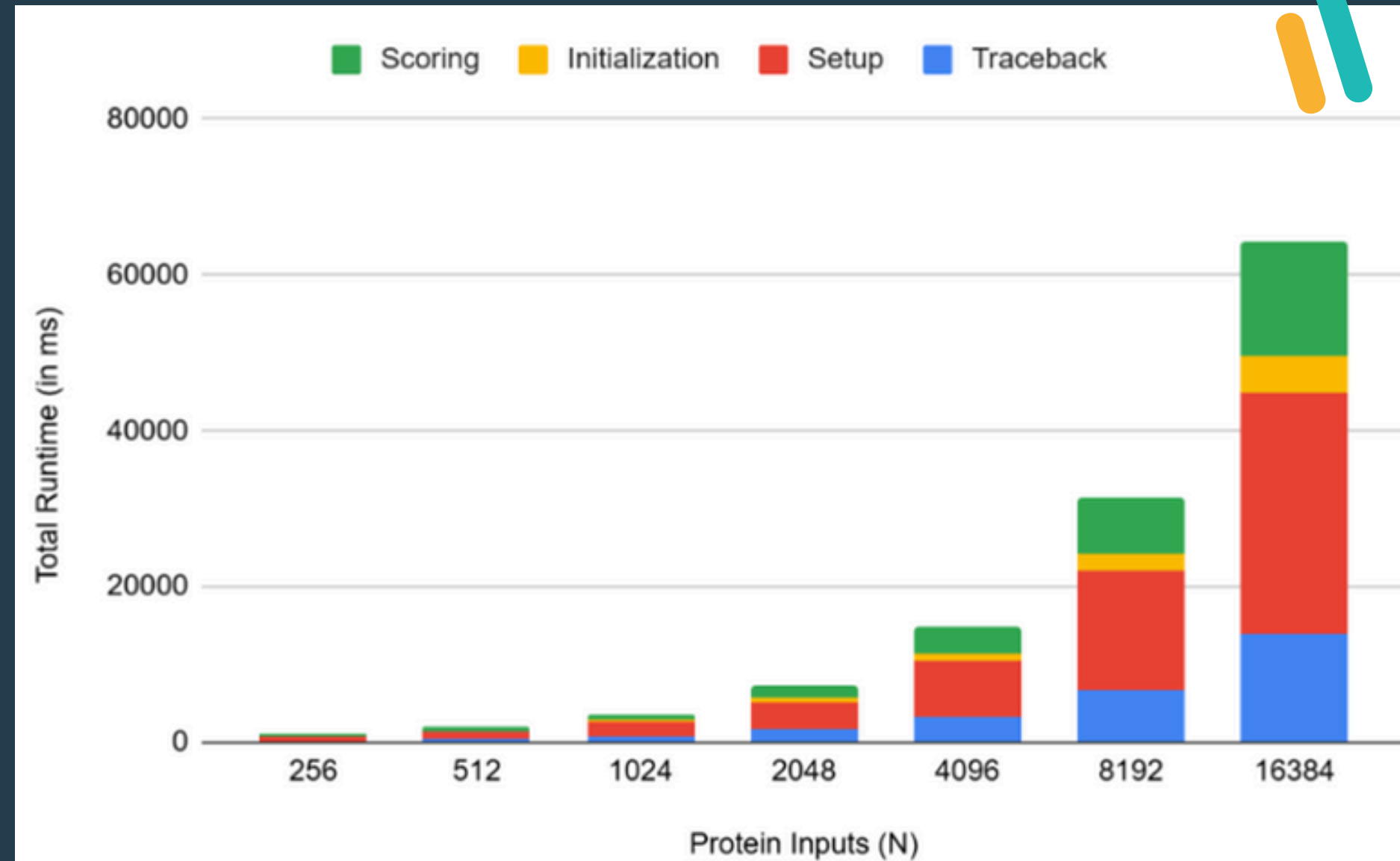


Figure 6-58: Scalability of the Runtime Contribution of Each Phase (Parallel)



Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F (Ox)

SCALABILITY TEST: INPUT

	Setup (in ms)	Initialization (in ms)	Scoring (in ms)	Traceback (in ms)	Total
256	3.8860 (0.30%)	92.3057 (7.12%)	1041.9393 (80.42%)	157.4784 (12.15%)	1295.6095
512	6.5797 (0.30%)	148.7886 (6.84%)	1750.4340 (80.44%)	270.2063 (12.42%)	2176.0086
1024	12.1077 (0.29%)	266.9724 (6.37%)	3384.3033 (80.74%)	528.3830 (12.61%)	4191.7664
2048	23.4261 (0.28%)	513.5938 (6.11%)	6803.7110 (80.94%)	1065.6262 (12.68%)	8406.3572
4096	48.4944 (0.28%)	1082.7895 (6.26%)	13995.0016 (80.88%)	2177.4801 (12.58%)	17303.7656
8192	102.3685 (0.28%)	2372.0643 (6.41%)	29925.7021 (80.85%)	4614.8657 (12.47%)	37015.0007
16384	201.0066 (0.26%)	4836.5212 (6.34%)	61761.0711 (80.96%)	9486.8561 (12.44%)	76285.4550

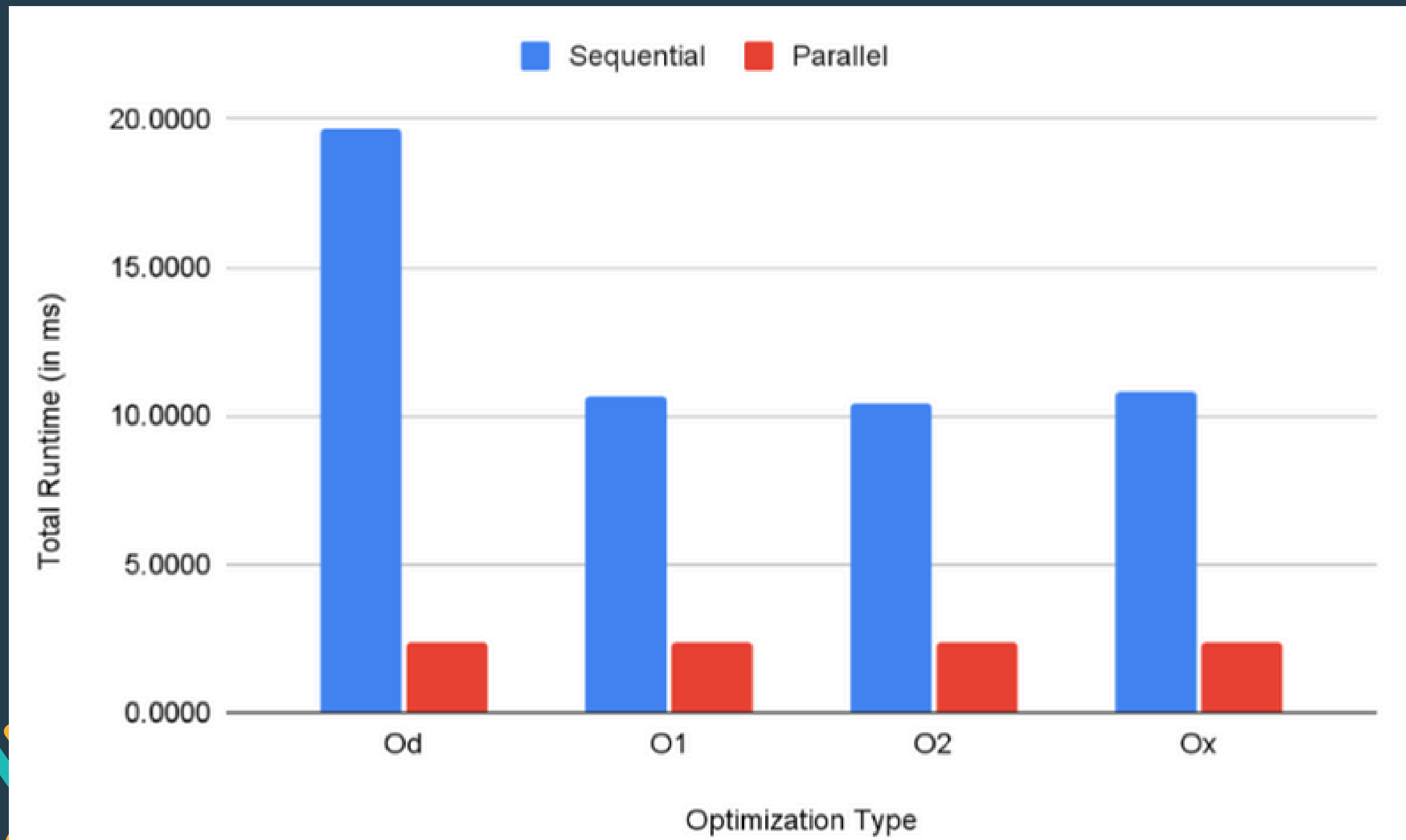
Table 6-12: Summary of Total Runtime Contribution of Each Phase (Sequential)

	Setup (in ms)	Initialization (in ms)	Scoring (in ms)	Traceback (in ms)	Total
256	508.4128 (47.34%)	92.8346 (8.64%)	242.4676 (22.58%)	230.1847 (21.43%)	1073.8998
512	910.3084 (48.11%)	150.0760 (7.93%)	428.8005 (22.66%)	402.8938 (21.29%)	1892.0788
1024	1771.5835 (48.19%)	268.3095 (7.30%)	844.4160 (22.97%)	791.5481 (21.53%)	3675.8571
2048	3541.3975 (48.20%)	509.8643 (6.94%)	1701.1427 (23.15%)	1595.3716 (21.71%)	7347.7762
4096	7211.7521 (48.09%)	1074.5773 (7.17%)	3457.5802 (23.06%)	3251.9264 (21.69%)	14995.8360
8192	15059.1698 (47.79%)	2364.3335 (7.50%)	7243.2962 (22.99%)	6844.1329 (21.72%)	31510.9325
16384	30850.5456 (47.86%)	4804.9875 (7.45%)	14812.6448 (22.98%)	13989.0137 (21.70%)	64457.1915

Table 6-13: Summary of Total Runtime Contribution of Each Phase (Parallel)

- Proportions of the program segments are consistent as dataset size grows

SCALABILITY TEST: COMPILER OPT

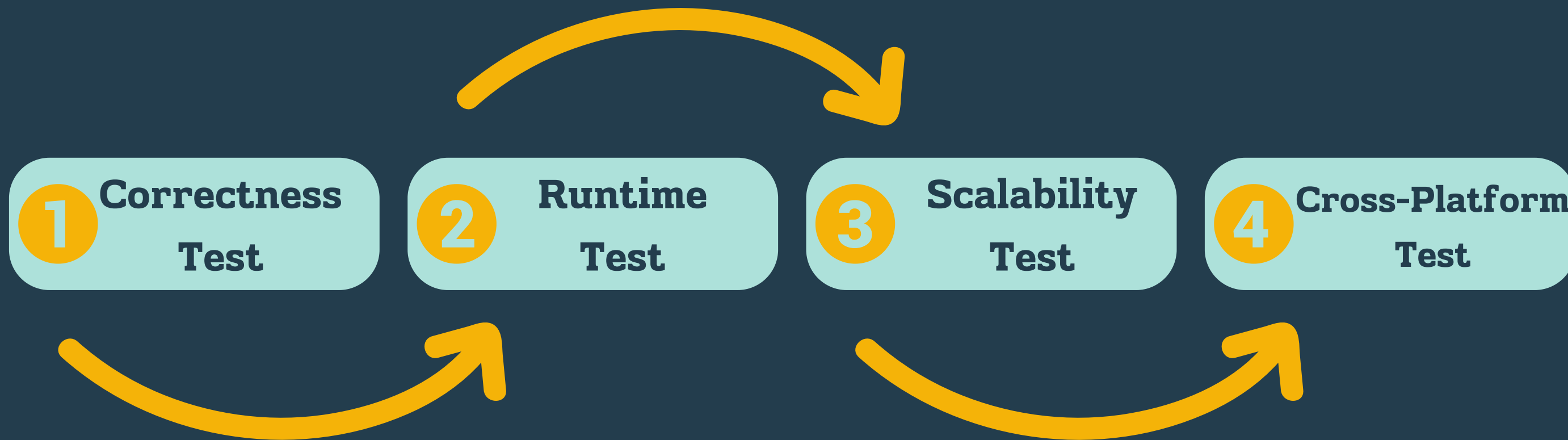


- **Minimal impact on the CUDA implementation**

Tested on desktop with an NVIDIA RTX 4070 Super GPU and an Intel Core i7-12700F

RESULTS & DISCUSSION

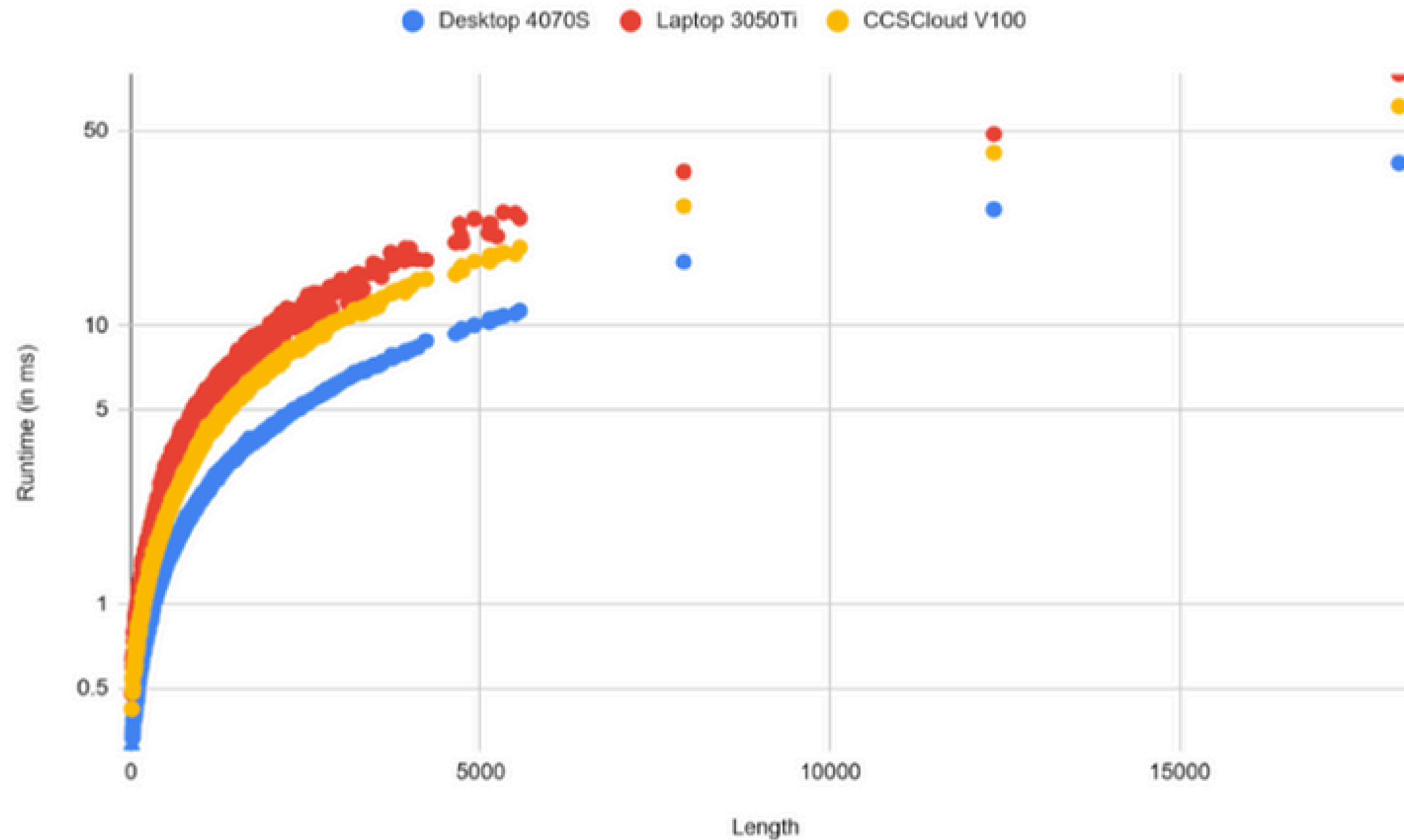
Upon completion of development, the implementation will be subjected to a series of tests. These evaluations will focus on ensuring **correctness**, assessing **runtime performance**, verifying **scalability**, and analyzing **cross-platform performance**.



CROSS-PLATFORM PERFORMANCE TEST

- This test provides insights into how the implemented program performs **across different hardware platforms**.
- Three platforms were examined in this section, namely, the **desktop** with an Intel i7-12700F and NVIDIA RTX 4070 Super, the **laptop** with a Ryzen 7 5800H and NVIDIA RTX 3050 Ti Mobile, and the **CCSCloud** Jupyter Environment with an Intel Xeon Silver 4210 and NVIDIA Tesla V100 PCIe 32 GB.

CROSS-PLATFORM PERFORMANCE TEST



- Desktop provides best performance, followed by CCSCloud and then Laptop

Figure 6-60: Parallel Implementation Runtime Across GPU Platforms

CROSS-PLATFORM PERFORMANCE TEST

	Laptop 3050Ti	CCSCloud VM0	Desktop 4070 Super
Average	1.7544s	8.5373s	3.5212s
Min	0.3225s	1.1129s	0.5468s
Max	3.2502s	12.4351s	5.9915s

	Laptop 3050Ti (in ms)		CCSCloud VM0 (in ms)		Desktop 4070 Super (in ms)	
	Sequential	SIMT	Sequential	SIMT	Sequential	SIMT
Average	11.1005	5.0417	37.4361	3.7952	10.8321	2.3915
Min	0.1619	0.4788	0.4767	0.421	0.1638	0.2983
Max	209.7535	80.1586	744.9806	61.4128	204.0286	38.4254

Table 6-16c Runtime Comparison Across GPU Platforms

CROSS-PLATFORM PERFORMANCE TEST

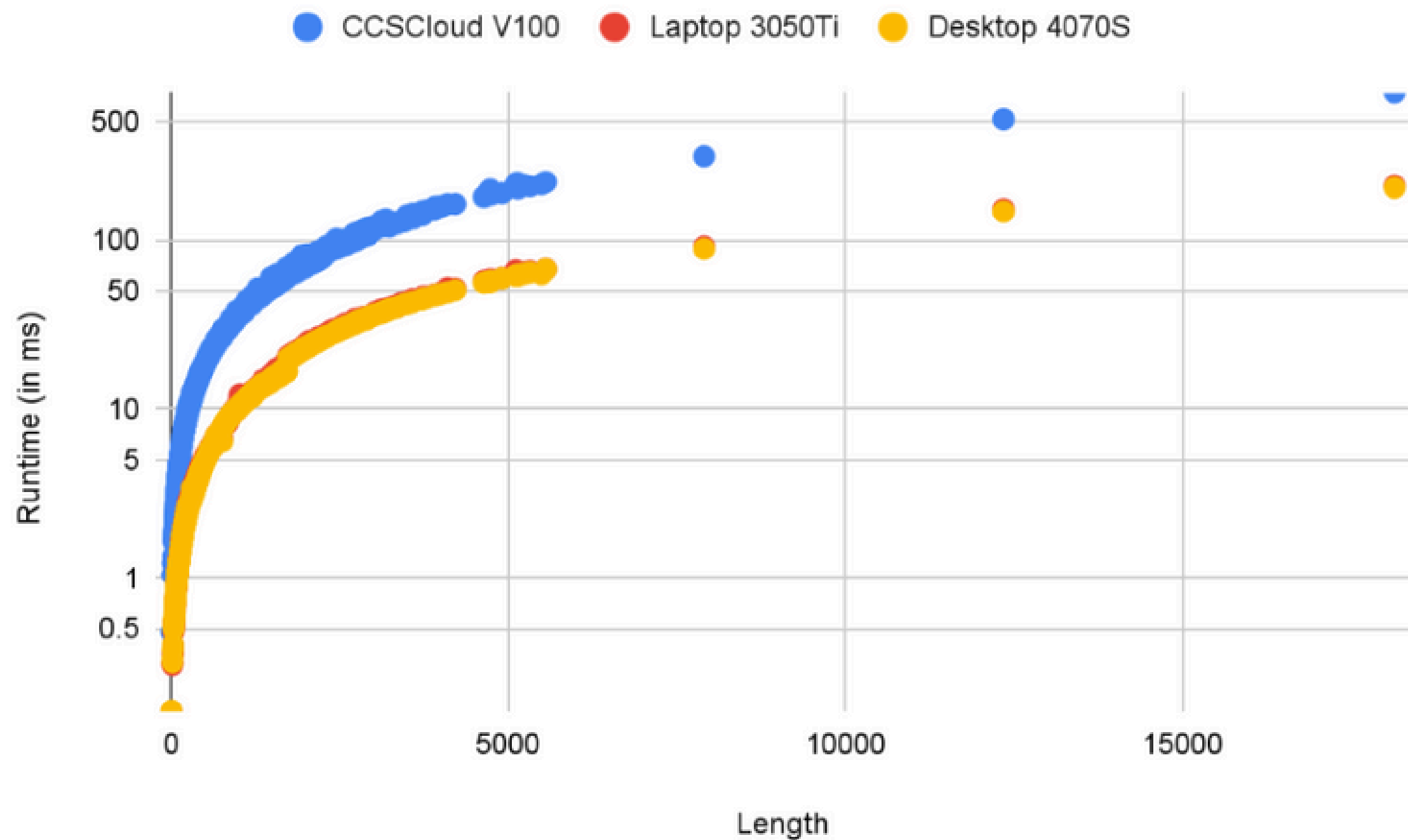


Figure 6-61: Sequential Implementation Runtime Across GPU Platforms

- 4070 Super < Tesla V100 < 3050ti Mobile

SPECIFIC OBJECTIVES



Developed a sequential implementation of Zhang's Six-Frame Alignment Algorithm.



Created an implementation scheme for Zhang's Six-Frame Alignment Algorithm to better fit the parallel framework of the SIMT model.



Implemented Zhang's Six-Frame Alignment Algorithm with the SIMT execution model using CUDA.



Evaluated the developed parallel implementation by comparing and contrasting its performance to the sequential version and existing SIMD implementation.



CONCLUSION & RECOMMENDATIONS

CONCLUSION

- **Demonstrated SIMT Acceleration:** Utilized the CUDA SIMT execution model to reduce computational costs and accelerate Zhang's Six-Frame Algorithm.
- **Parallelization:** Implemented a hybrid multi-level anti-diagonal wavefront technique, adapted for the SIMT model, proving the viability of this paradigm in CUDA.
- **Significant Speedups:** Achieved $\sim 3.5x$ speedup over Sequential implementation. Also achieved $\sim 3.1x$ speedup with SIMD (AVX2).



CONCLUSION

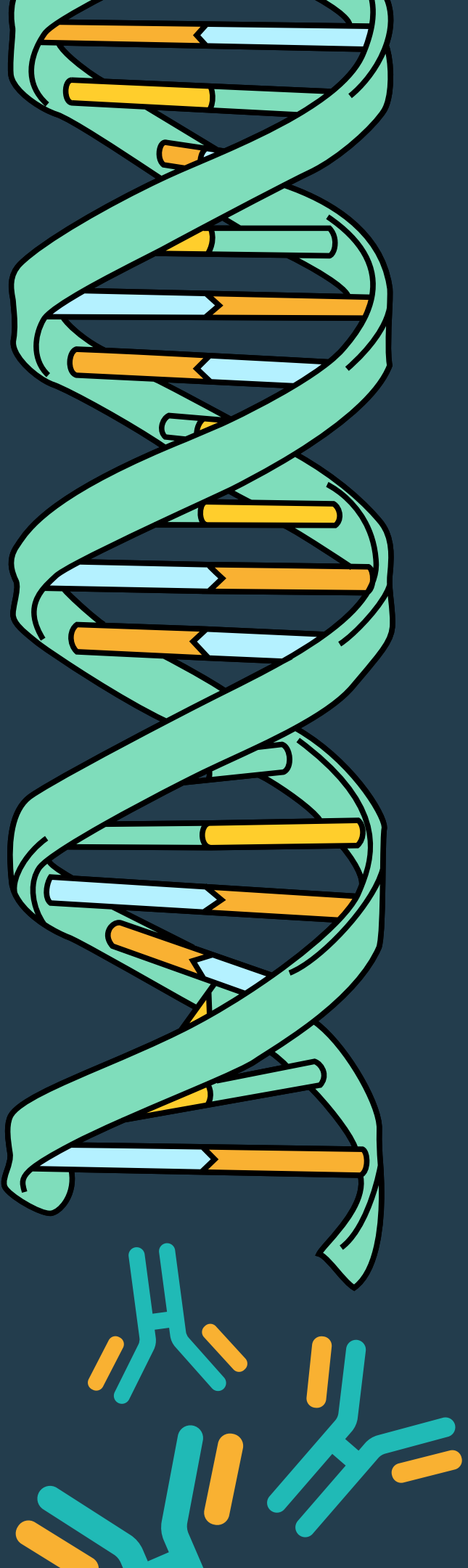
- **Hardware Impact:** Showcased the impact of GPU hardware quality on CUDA code performance.
 - **Compiler Impact:** Compiler optimizations had minimal effect on CUDA performance (due to GPU-specific compilation and architecture factors), but showed considerable improvement for the Sequential CPU implementation.
 - **Proof of Concept:** Validates the ability of GPUs for complex bioinformatics algorithms, enabling faster data processing and downstream genetic analysis.
- 

RECOMMENDATIONS

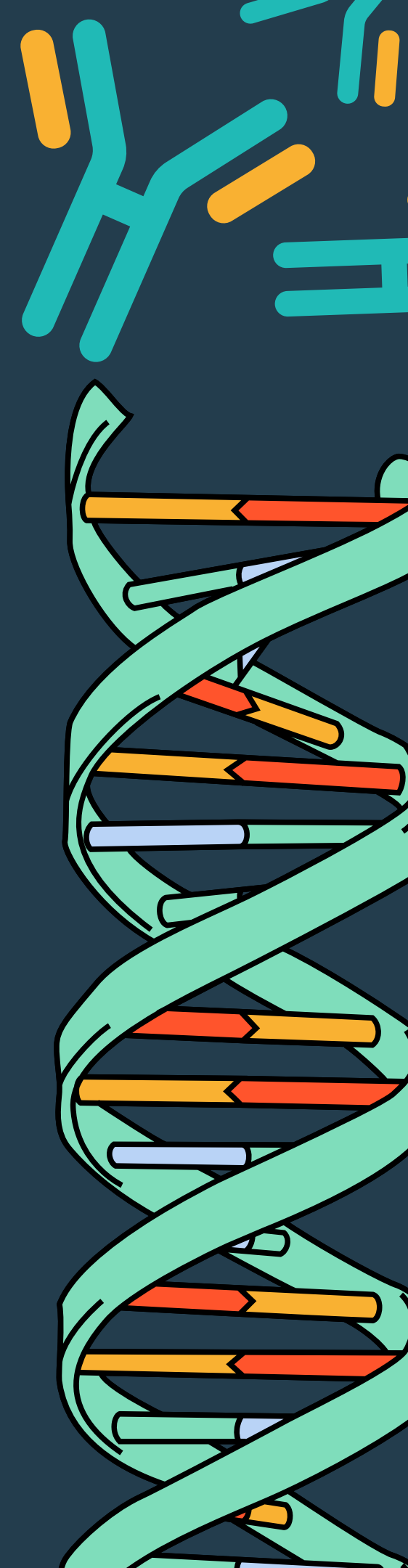
- **Implementation Optimization:** Use of a purely device implementation scheme can potentially enable better performance.
- **Holistic Optimization:** Exploring speedups in the other phases of the program, including performing initialization using device code, as well as in the setup phase, particularly with regard to data transfers between the device and the host.

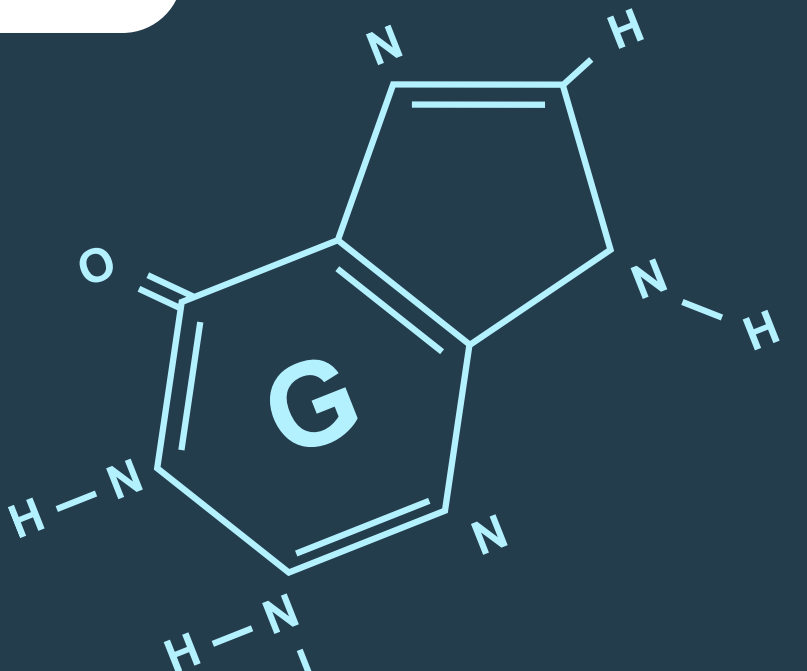
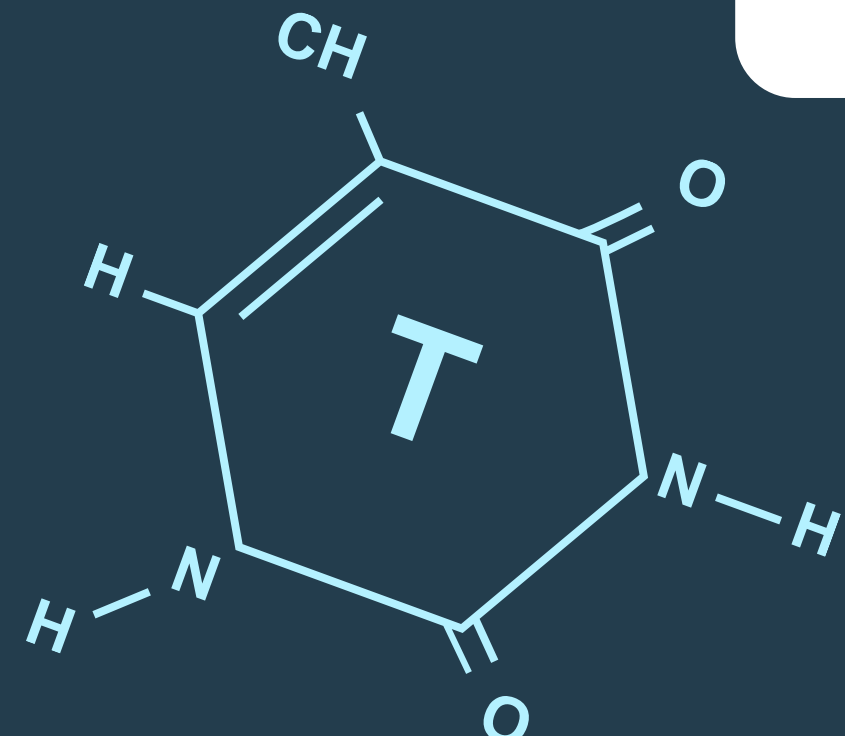
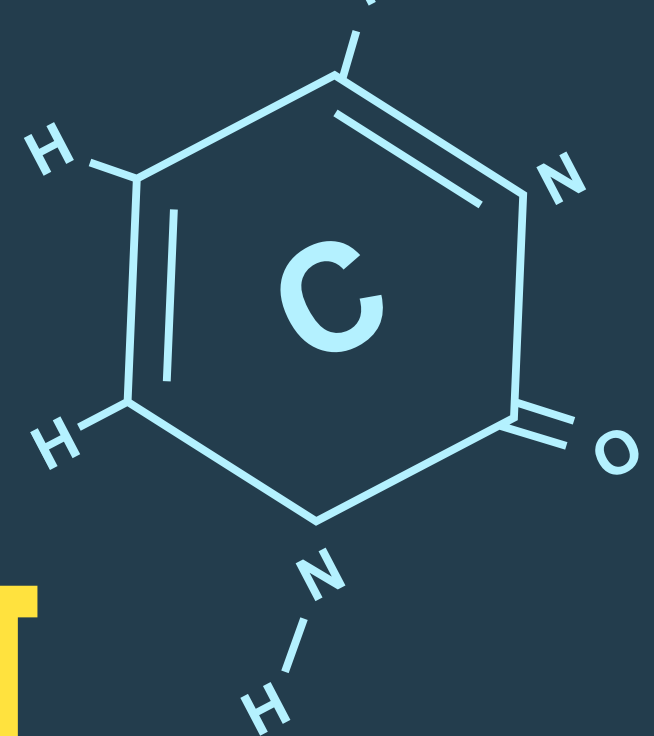
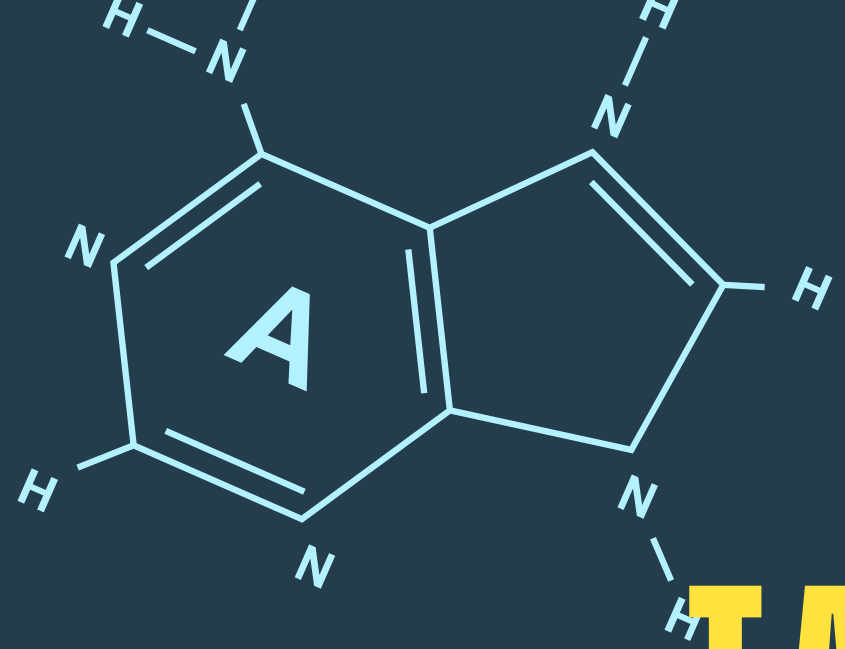
RECOMMENDATIONS

- **Comprehensive Hardware Testing:** Some improvements to the testing methodology can also be made, particularly with regard to testing performance across multiple devices.
- **Utilize CUDA C/C++ Libraries:** To explore the use of libraries of CUDA C/C++ that can lessen the overhead/computation cost.



DEMO





TARA CUDA: A HYBRID SIMT IMPLEMENTATION OF ZHANG'S SIX- FRAME ALGORITHM USING CUDA

ARCETA - MENDOZA - TAN

ADVISER: MR. ROGER LUIS UY